

innovate

achieve

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BITS Pilani

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SURVEYING (CE F213)

(Curves)

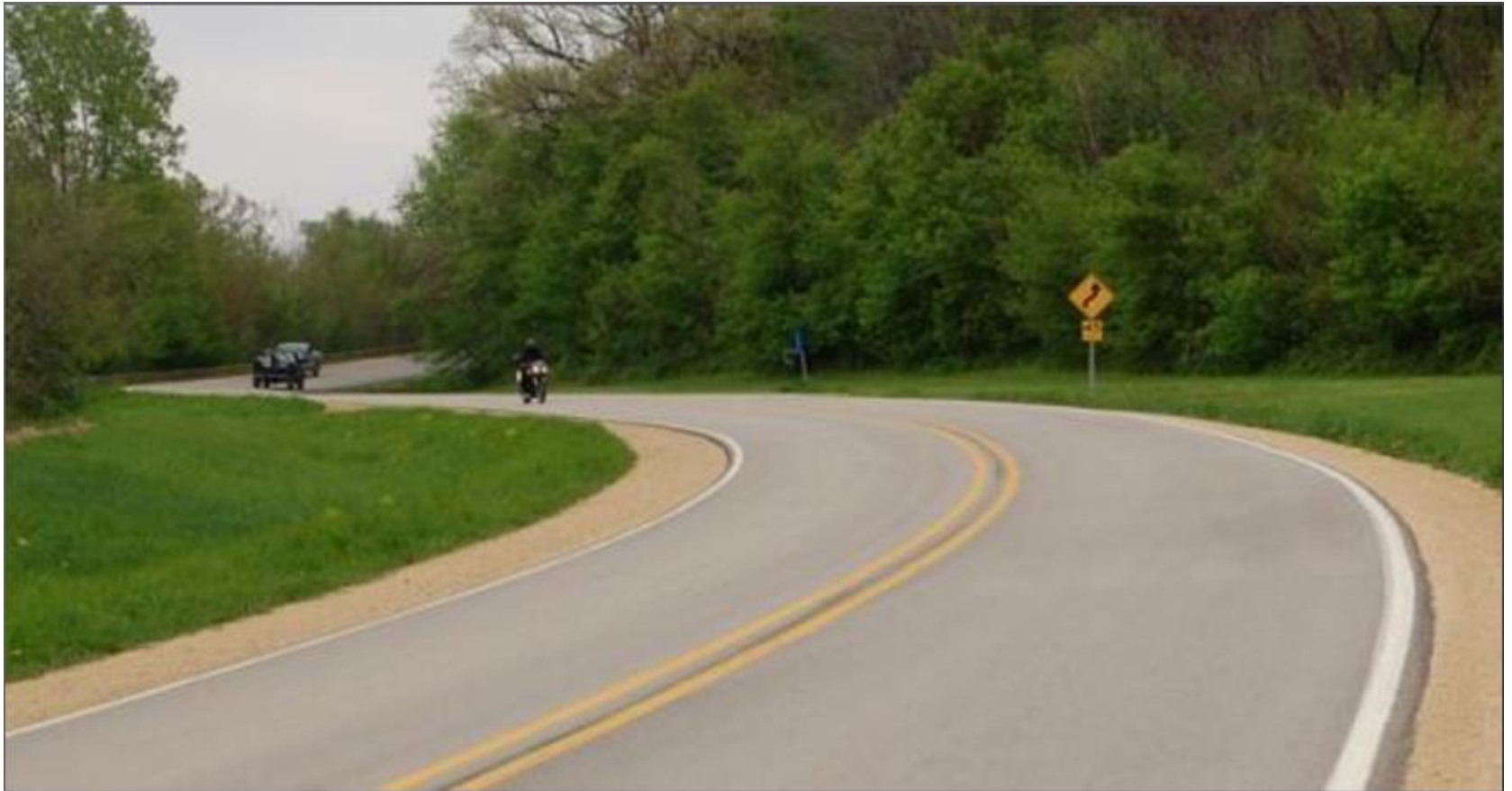
Curves



Necessity of Curves

1. Change in alignment – not in one straight line
2. Change in direction
3. Change in slope of the roads
4. If costly land comes along the straight line, curves can be used to avoid that patch
5. If obstruction comes along the line
6. Cutting and filling can be minimized by using curves
7. Generally used on Highway and Railway

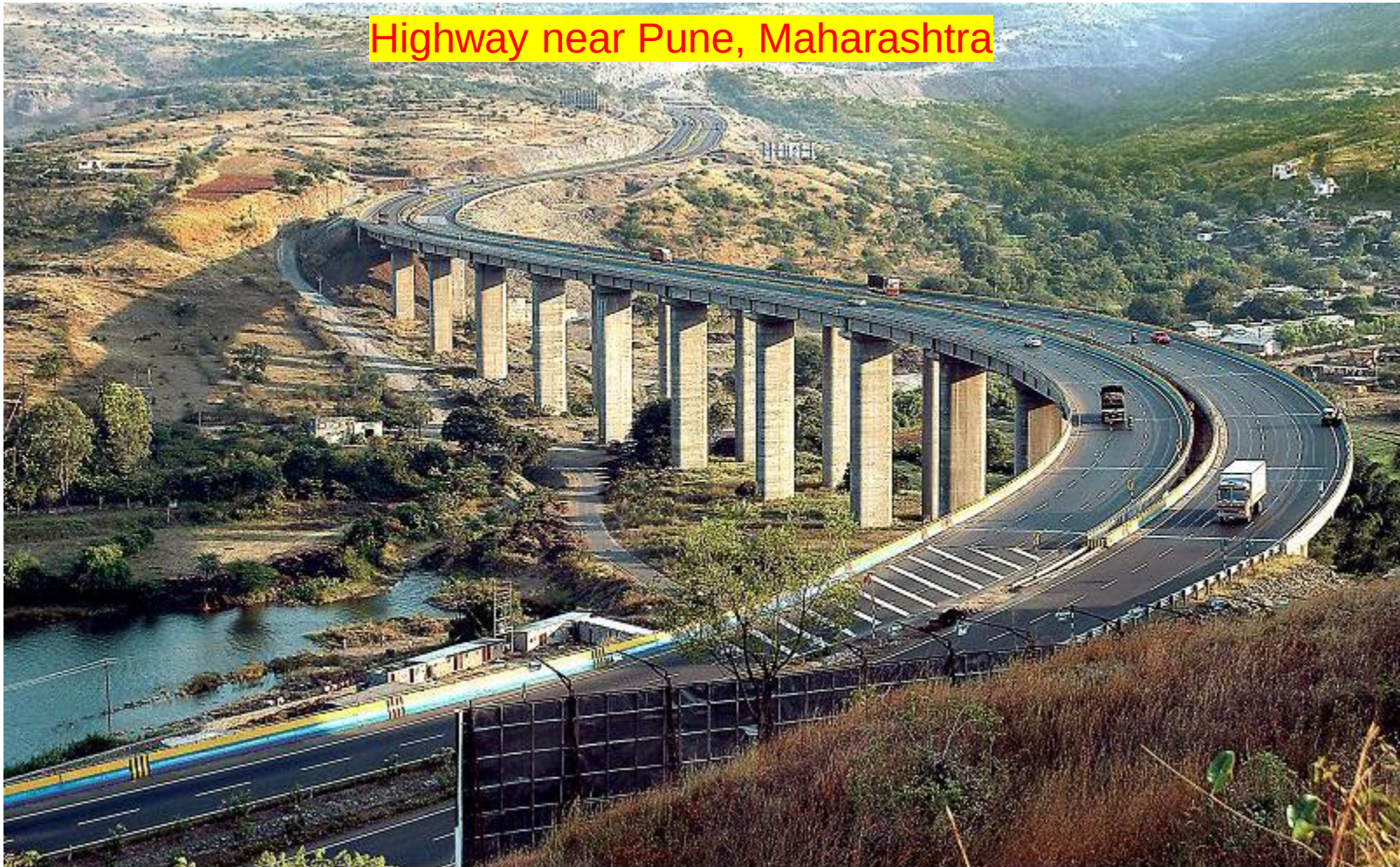
A curve is a line that gradually and smoothly bends or turns, without any sharp turns or abrupt turns.





Spiral curves have been featured in highway and development designs in addition to railroads. There was a mini-boom in the use of spirals during the postwar improvements to the nation's scenic parkways—*courtesy of the National Park Service.*

Highway near Pune, Maharashtra



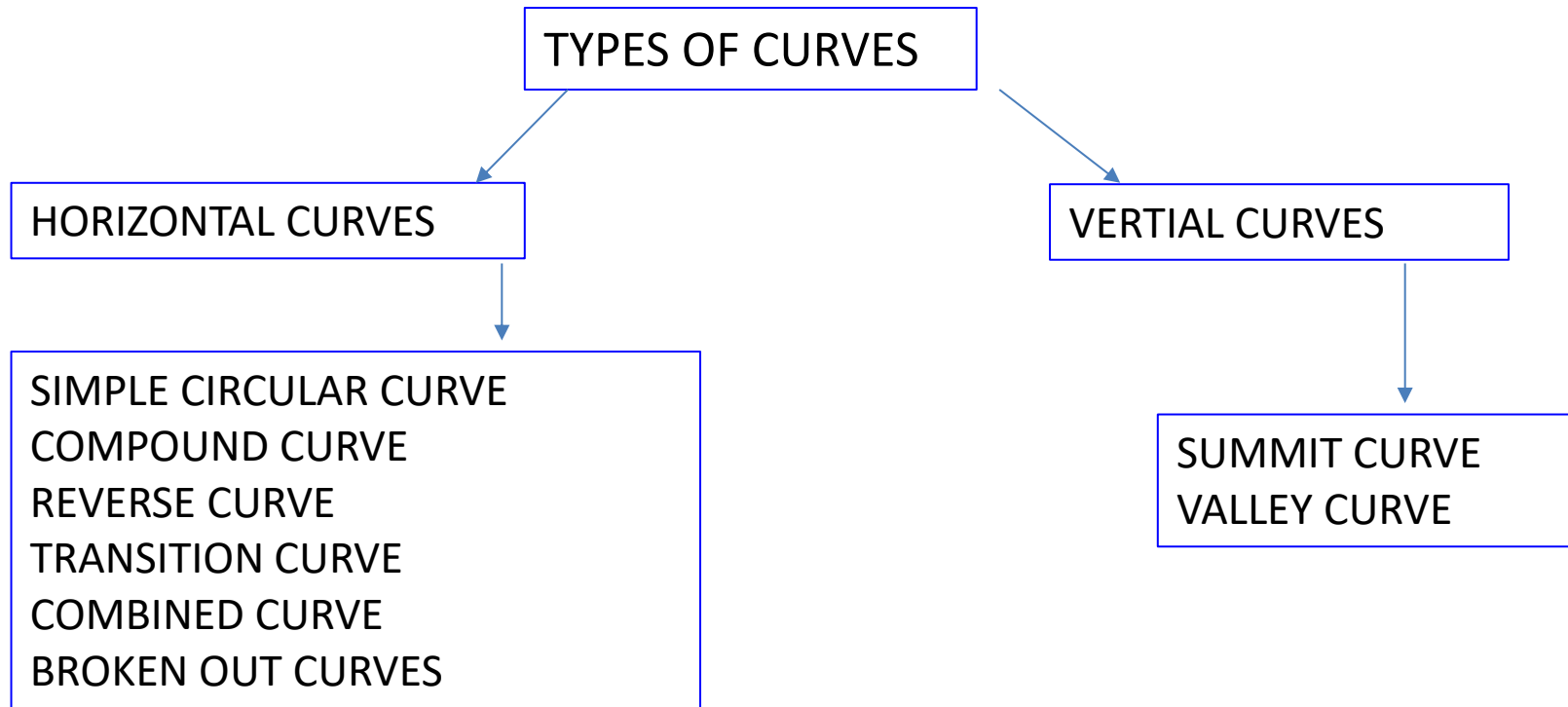






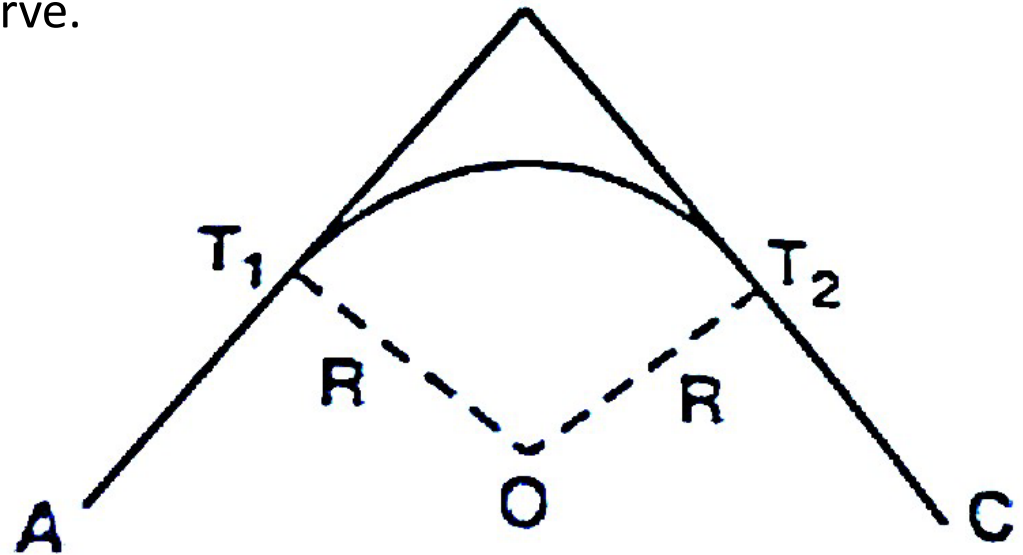






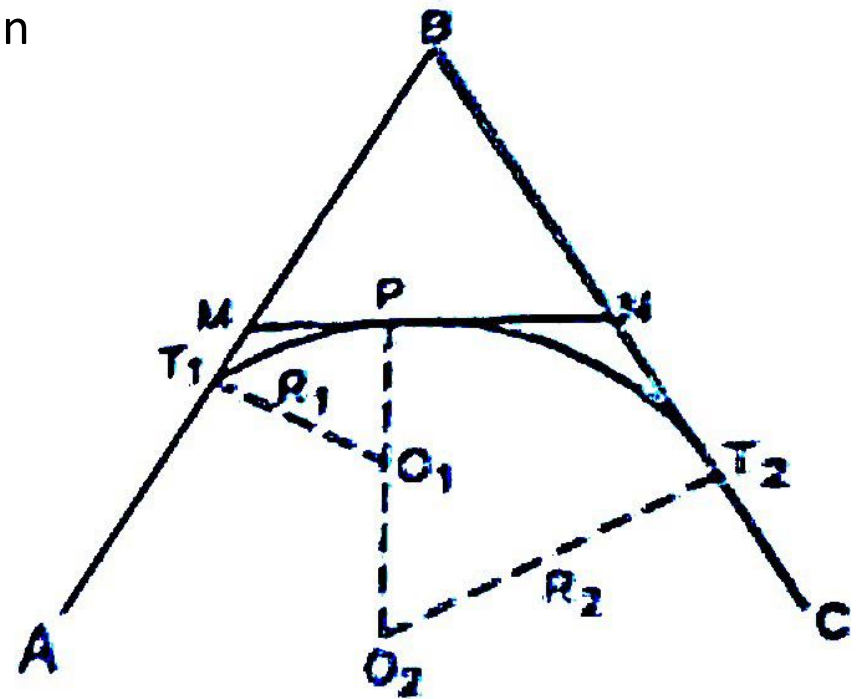
SIMPLE CIRCULAR CURVE

A simple curve consist of a single arc connecting two straights or tangents. Simple curve is normally represented by the length of its radius or by the degree of curve.



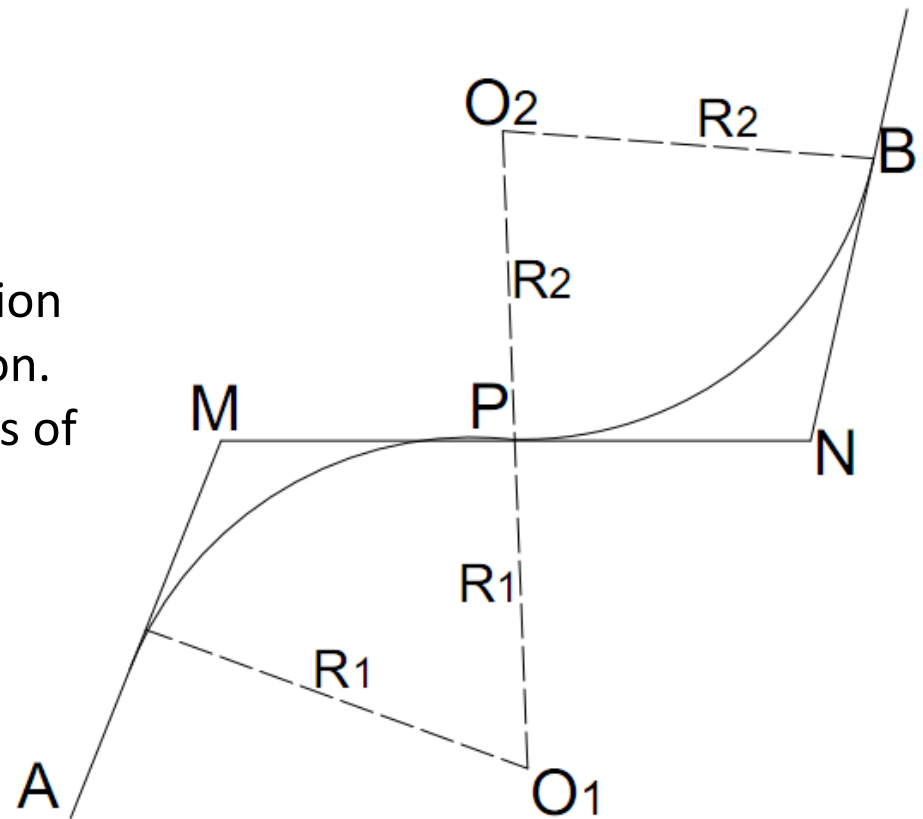
COMPOUND CURVE

A compound curve consist of two arcs of different radii curving in the same direction and lying on the same side of their common tangent. Their centers remain on the same side of the curve.



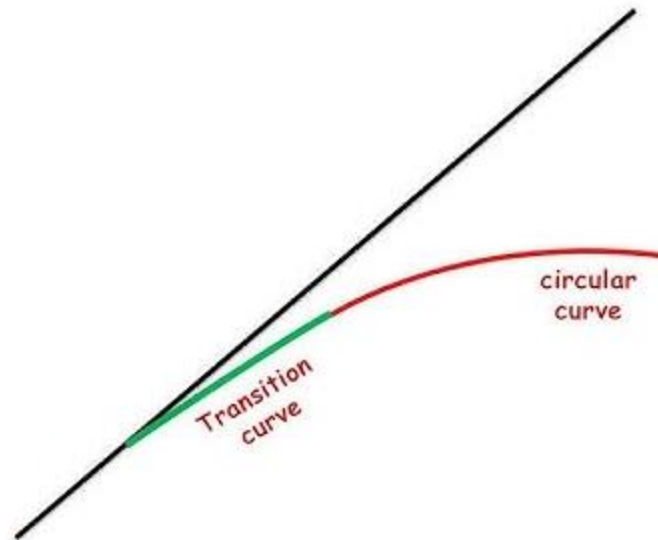
REVERSE CURVE

A reverse curve is composed of two arcs of equal or different radii bending or curving in opposite direction with common tangent at their junction. Their centers remain in opposite sides of the curve.



TRANSITION CURVE

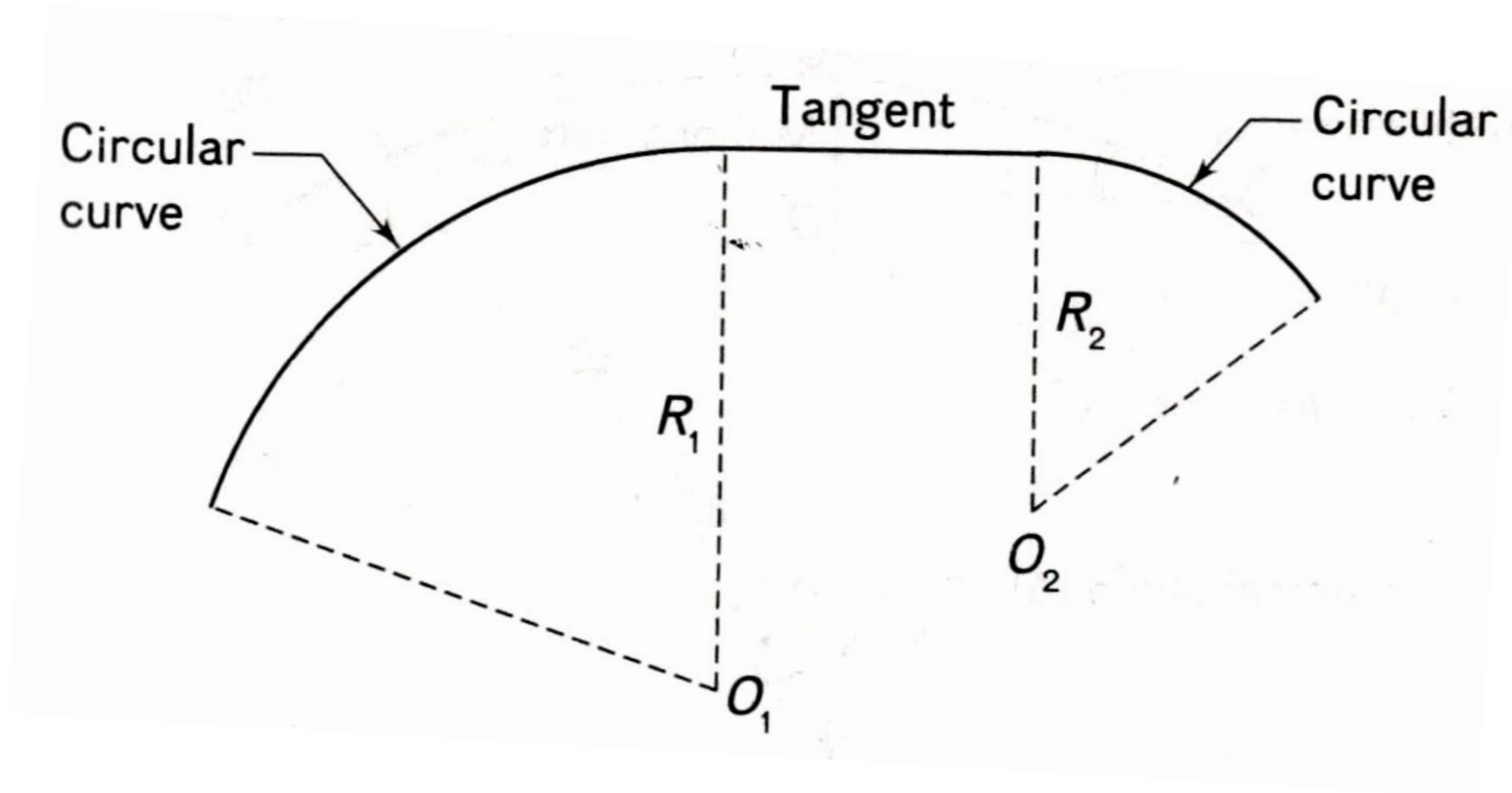
It is the curve between straight line and simple circular curve.



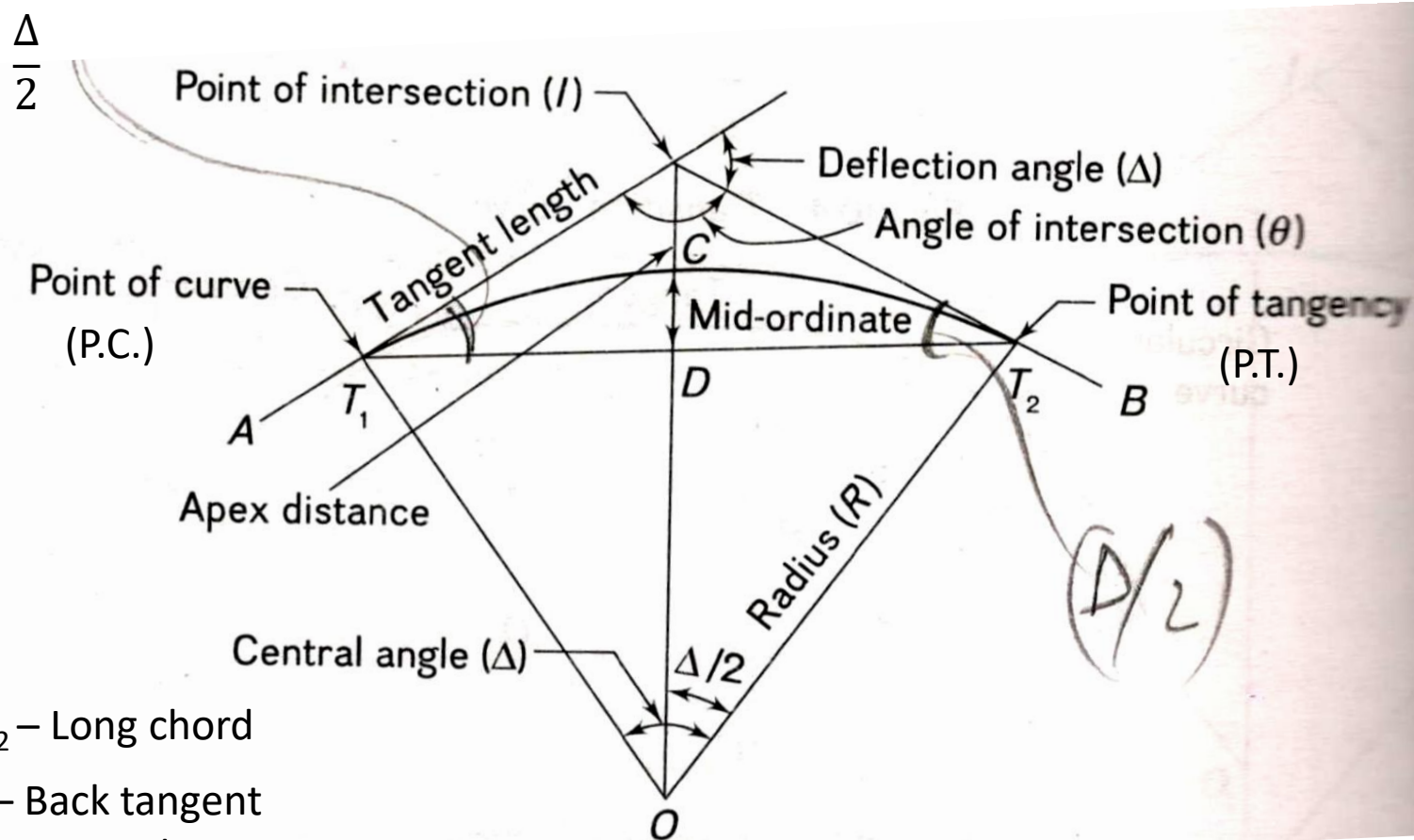
It is the curve having the combination of transition curves and circular curve.



BROKEN OUT CURVES



Elements of simple circular curve



T_1T_2 – Long chord

IT_1 – Back tangent

IT_2 – Forward tangent

Degree of Curve

The degree of curve can be defined in two ways:

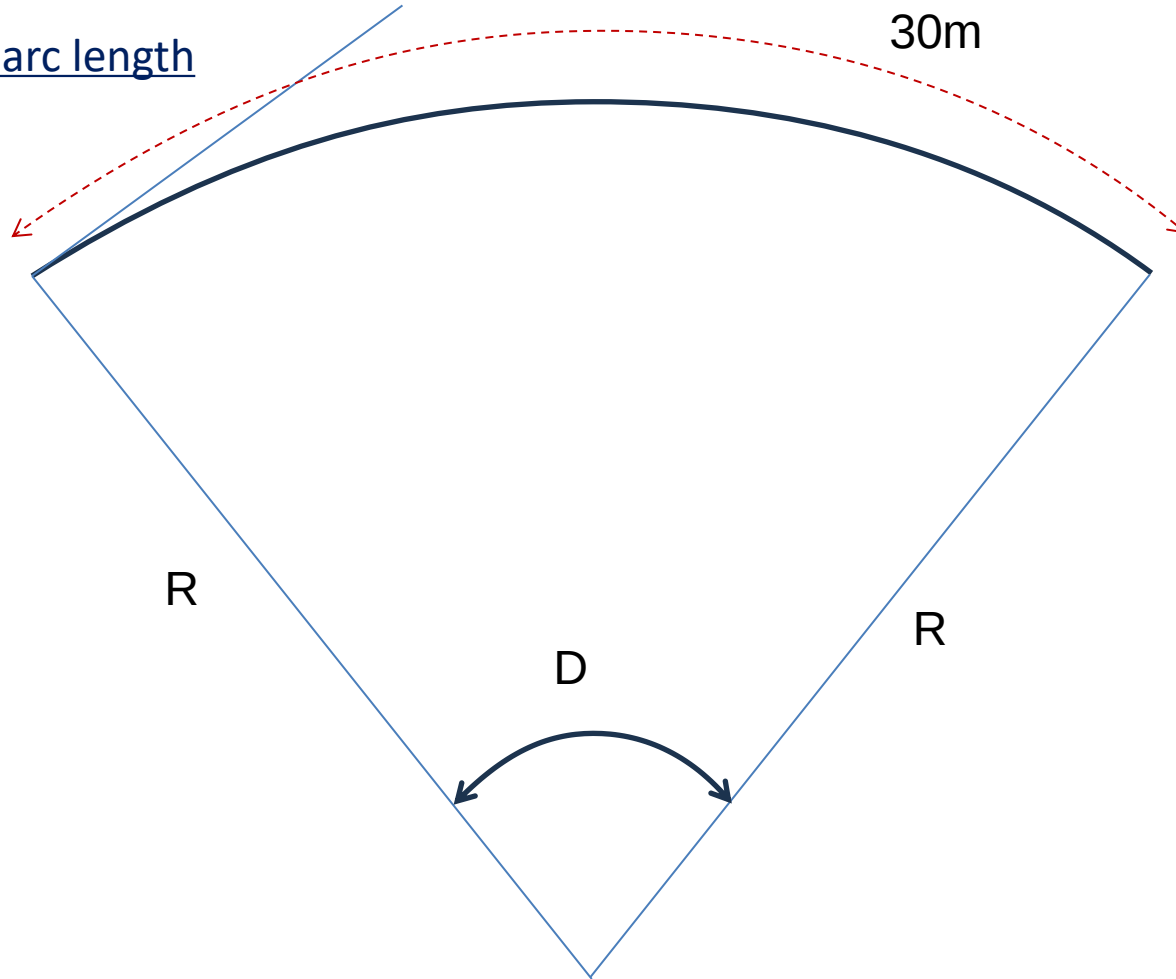
- (i) Arc definition
- (ii) Chord definition

Arc definition

Degree of curve is the central angle subtended by a fixed length of Arc. The length of the arc is normally 20m or 30m.

So the degree of curve should be specified with respect to 20m or 30m length.

For 30m arc length



$$R \cdot D = 30 \quad (D \text{ in radian})$$

$$R \left(D * \frac{\pi}{180} \right) = 30 \quad (D \text{ in degree})$$

$$R = \frac{30 * 180}{\pi D} \quad (D \text{ in degree})$$

$$R = \frac{1719}{D} \quad (D \text{ in degree})$$

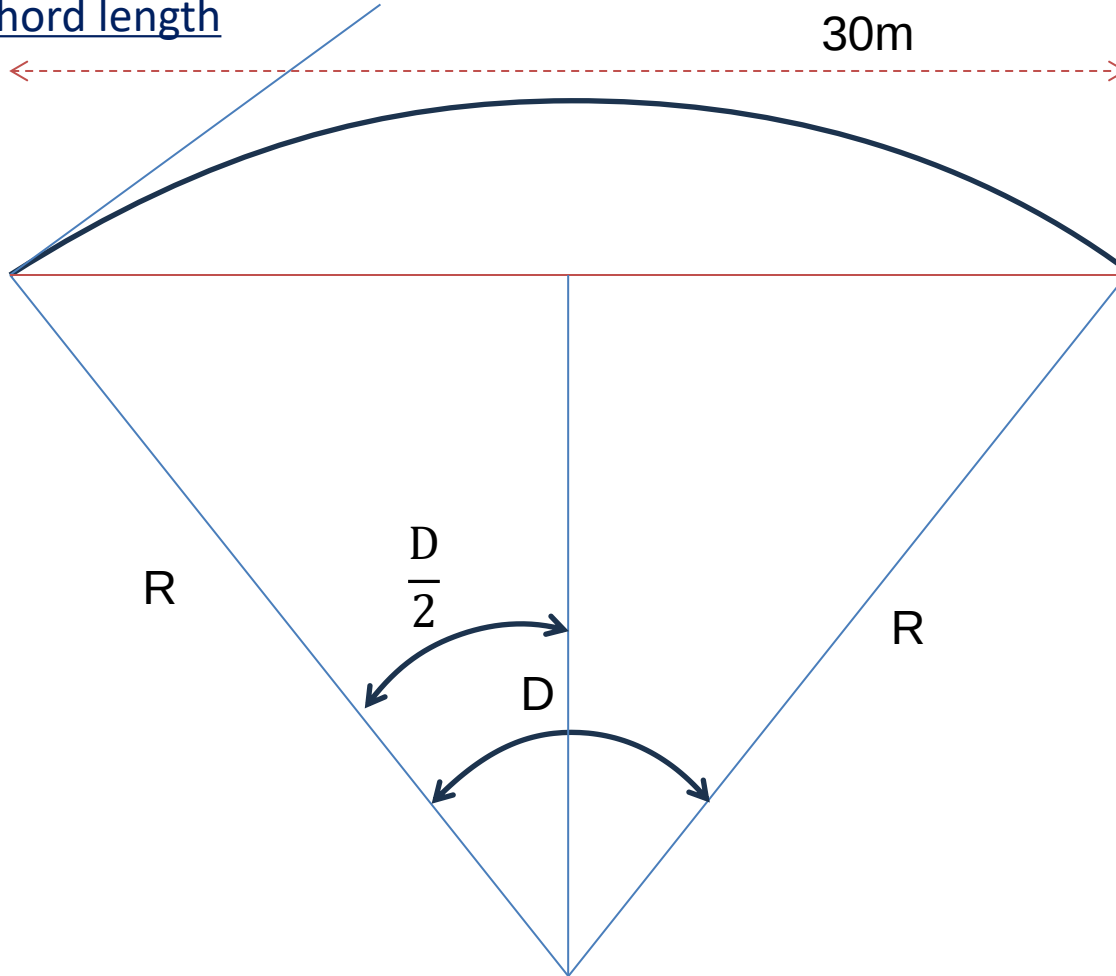
Similarly for 20m arc length

$$R = \frac{1146}{D} \quad (D \text{ in degree})$$

Chord definition

Degree of curve is the central angle subtended by a fixed length of Chord. The length of the Chord is normally 20m or 30m.

For 30m chord length



$$\sin \frac{D}{2} = \frac{15}{R} \quad (D \text{ in radian})$$

$$\frac{D}{2} = \frac{15}{R} \quad (D \text{ in radian}), \text{ considering } D \text{ to be small.}$$

$$R \cdot D = 30 \quad (D \text{ in radian})$$

$$R \left(D * \frac{\pi}{180} \right) = 30 \quad (D \text{ in degree})$$

$$R = \frac{30 * 180}{\pi D} \quad (D \text{ in degree})$$

$$R = \frac{1719}{D} \quad (D \text{ in degree})$$

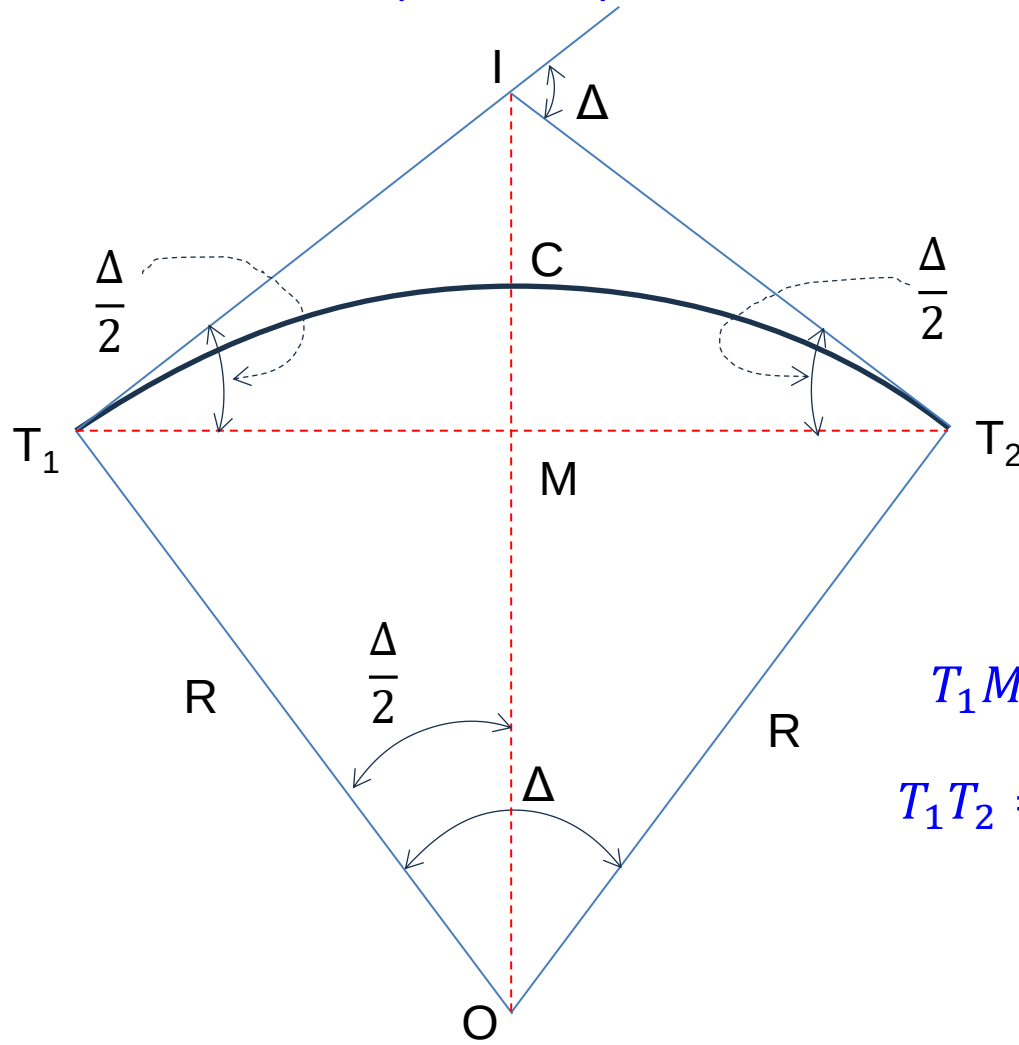
Similarly for 20m chord length

$$R = \frac{1146}{D} \quad (D \text{ in degree})$$

Degree of curve is the relationship between the Radius of curve and central angle subtended by a fixed length of Chord or fixed length of Arc.

The fixed length of the Chord or Arc is normally 20m or 30m. In both case we are getting same relation by considering D to be small.

Different mathematical relationships in simple circular curve



$$T_1M = MT_2 = \frac{L}{2}$$

$$T_1T_2 = L \text{ (long chord)}$$

(i) Tangent Length (IT_1)

In triangle IT_1O , we can write $IT_1 = R \tan \frac{\Delta}{2}$

In triangle IT_1M , we can write $IT_1 = \frac{\frac{L}{2}}{\cos \frac{\Delta}{2}}$

(ii) Long Chord Length (T_1T_2)

In triangle T_1MO , we can write $T_1M = \frac{L}{2} = R \sin \frac{\Delta}{2}$

So, $T_1T_2 = L = 2R \sin \frac{\Delta}{2}$

(iii) Mid Ordinate (CM)

$$CM = OC - OM = R - R \cos \frac{\Delta}{2} = R \left(1 - \cos \frac{\Delta}{2} \right)$$

(iv) Apex Distance (IC)

$$IC = IO - R$$

In triangle IT_1O , we can write, $\cos \frac{\Delta}{2} = \frac{R}{IO}$

$$\text{So, } IO = \frac{R}{\cos \frac{\Delta}{2}} = R \sec \frac{\Delta}{2}$$

$$IC = R \sec \frac{\Delta}{2} - R = R \left(\sec \frac{\Delta}{2} - 1 \right)$$

(v) Length of the Curve (T_1CT_2)

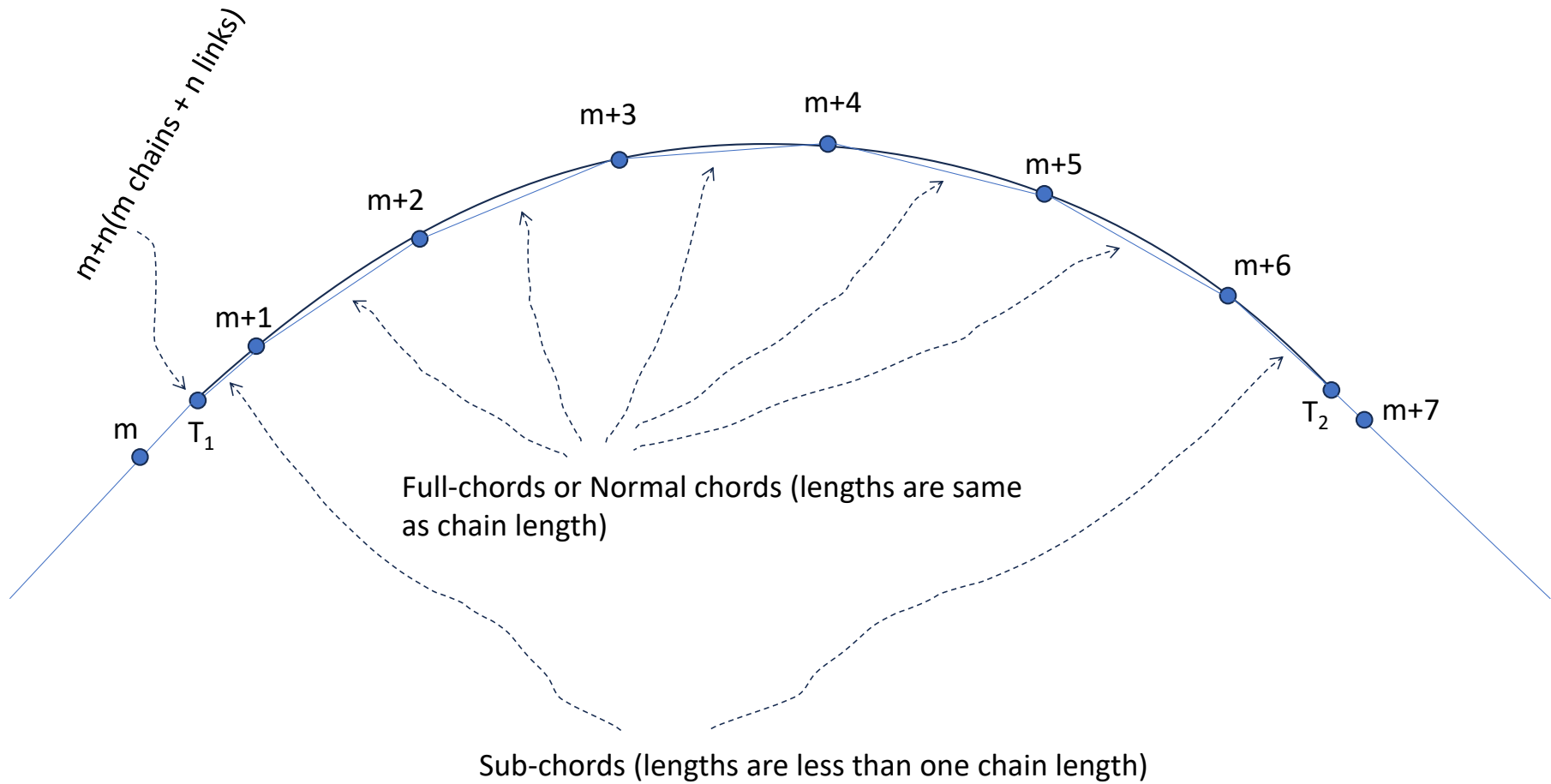
$$\text{Length of the Curve} = T_1CT_2 = R\Delta \quad (\Delta \text{ in Radian})$$

Peg interval, full or Normal chord and sub-chord

Peg interval : The curve length between any two successive pegs is called peg interval. However, it is not practical to measure the peg interval along the curve, it is measured along the chord.

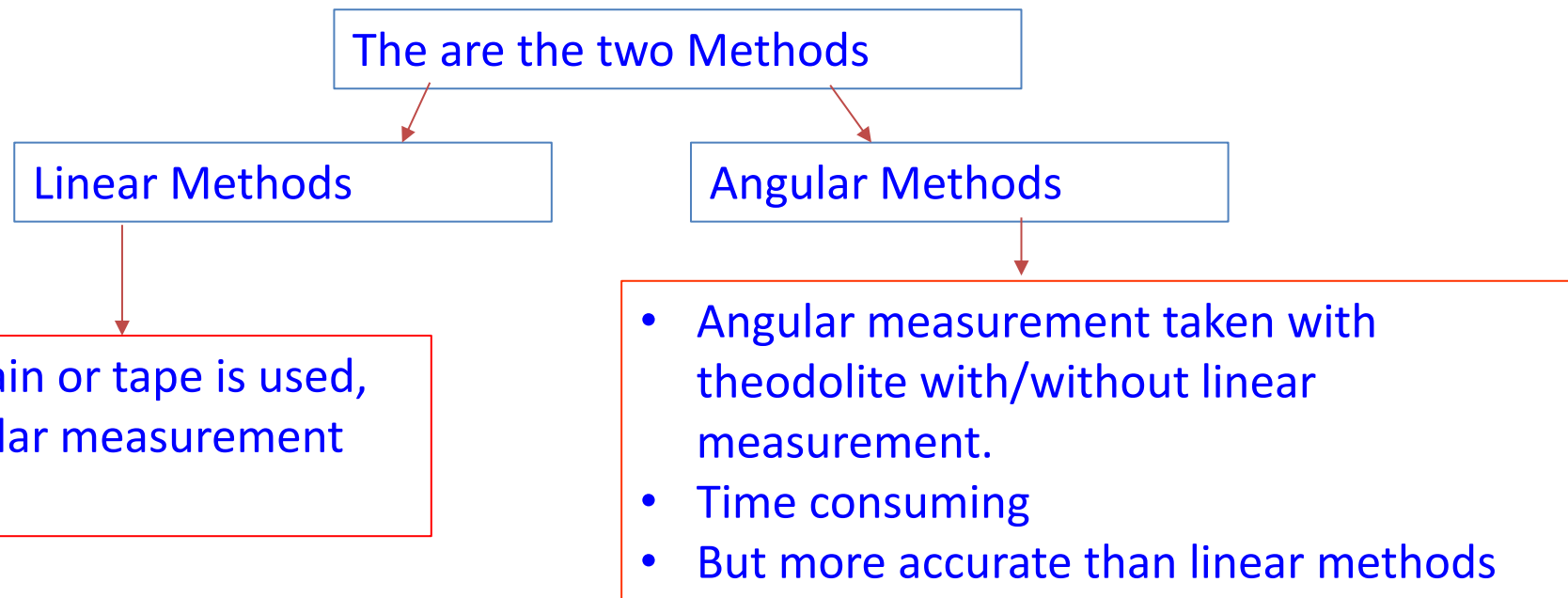
if *chord length* $\leq \frac{1}{20}$ of *radius of curve*, the chord length and the curve length becomes nearly equal.

The usual peg interval is 20m or 30m. It may be reduced in sharp curves.



Setting out of simple circular curve

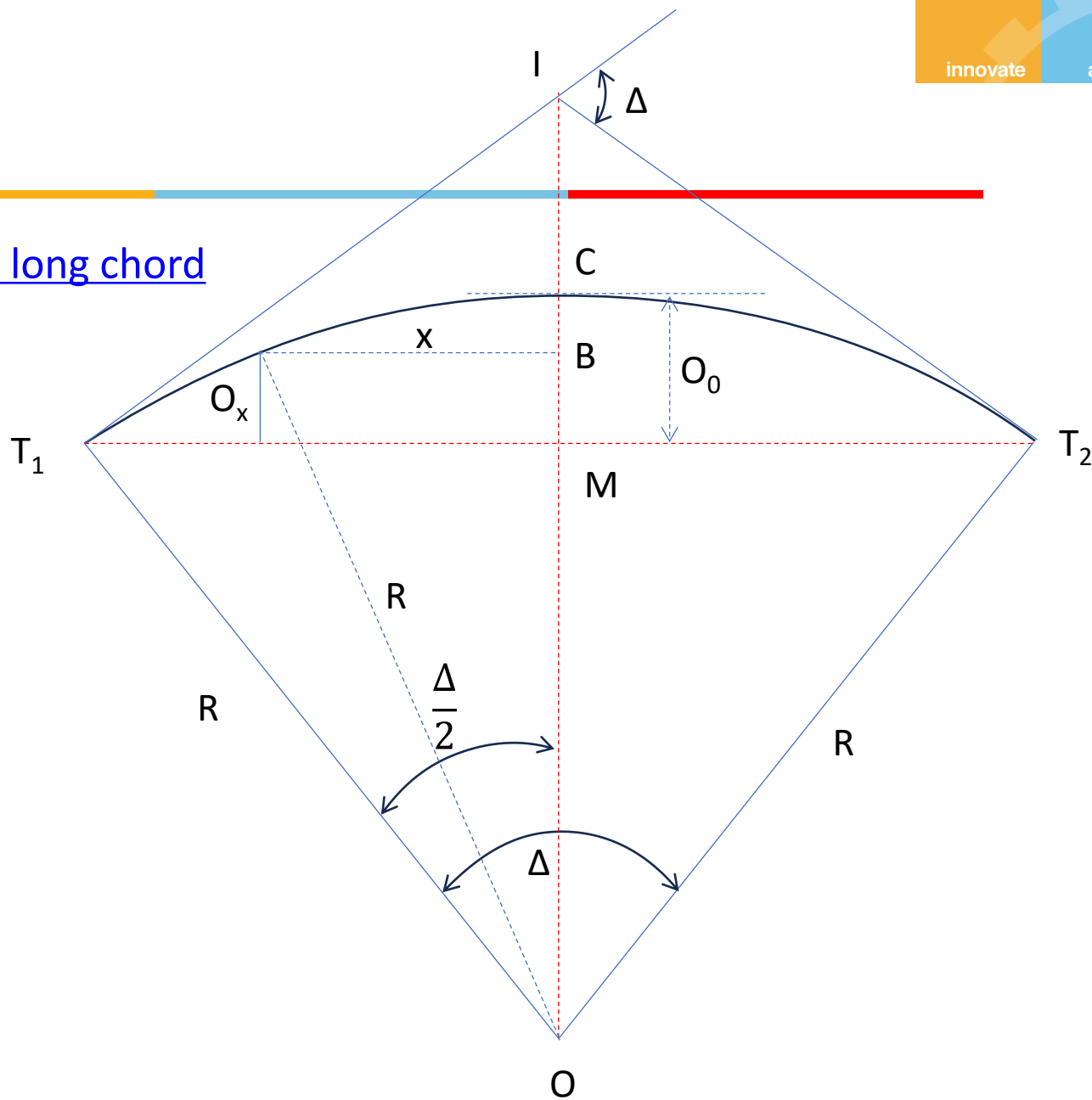
Setting out of curve means locating various points along the length of curve at convenient intervals.



Linear Methods

The various linear methods are,

- (i) Offset from long chord
- (ii) Perpendicular offset from tangent
- (iii) Radial Offset from tangent
- (iv) Successive bi-section of arcs or chords
- (v) Offset from Chord produced



$$O_x = OB - OM$$

$$O_x = OB - (OC - CM)$$

$$O_x = \sqrt{(R^2 - x^2)} - (R - O_0)$$

Exact Expression, (O_0) is mid ordinate

$$O_x = \sqrt{\frac{R^2(R^2 - x^2)}{R^2}} - (R - O_0)$$

$$O_x = R \sqrt{1 - \left(\frac{x}{R}\right)^2} - (R - O_0)$$

$$O_x = R \left(1 - \frac{1}{2} \left(\frac{x^2}{R^2} \right) \right) - (R - O_0)$$

Using binomial Expansion

$$O_x = O_0 - \frac{x^2}{2R}$$

Approximate Expression

$$O_0 = R - \sqrt{R^2 - \left(\frac{L}{2}\right)^2}$$

$$O_0 = R \left(1 - \cos \frac{\Delta}{2} \right)$$

$$O_x = R - OA$$

$$O_x = R - \sqrt{R^2 - x^2}, \text{ Exact Expression}$$

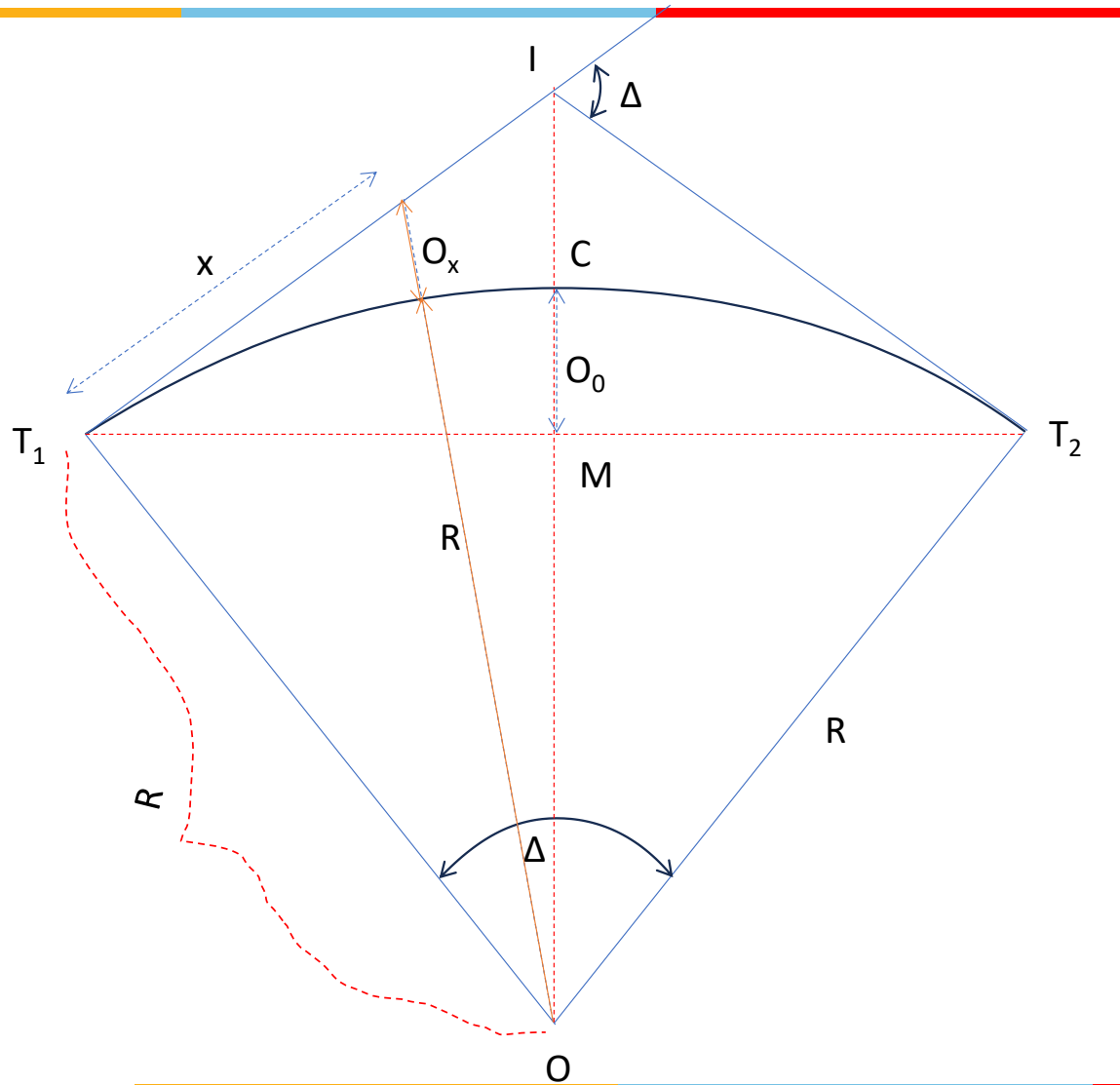
$$O_x = R - \sqrt{\frac{R^2(R^2 - x^2)}{R^2}}$$

$$O_x = R - R\sqrt{1 - \left(\frac{x}{R}\right)^2}$$

$$O_x = R - R\left(1 - \frac{1}{2}\left(\frac{x^2}{R^2}\right)\right), \text{ Using binomial Expansion}$$

$$O_x = \frac{x^2}{2R} \quad \text{Approximate Expression}$$

Radial Offset from tangent



$$(R + O_x)^2 = R^2 + x^2$$

$$R + O_x = \sqrt{R^2 + x^2}$$

$$O_x = \sqrt{R^2 + x^2} - R \quad \text{Exact Expression}$$

$$O_x = \sqrt{\frac{R^2(R^2 + x^2)}{R^2}} - R$$

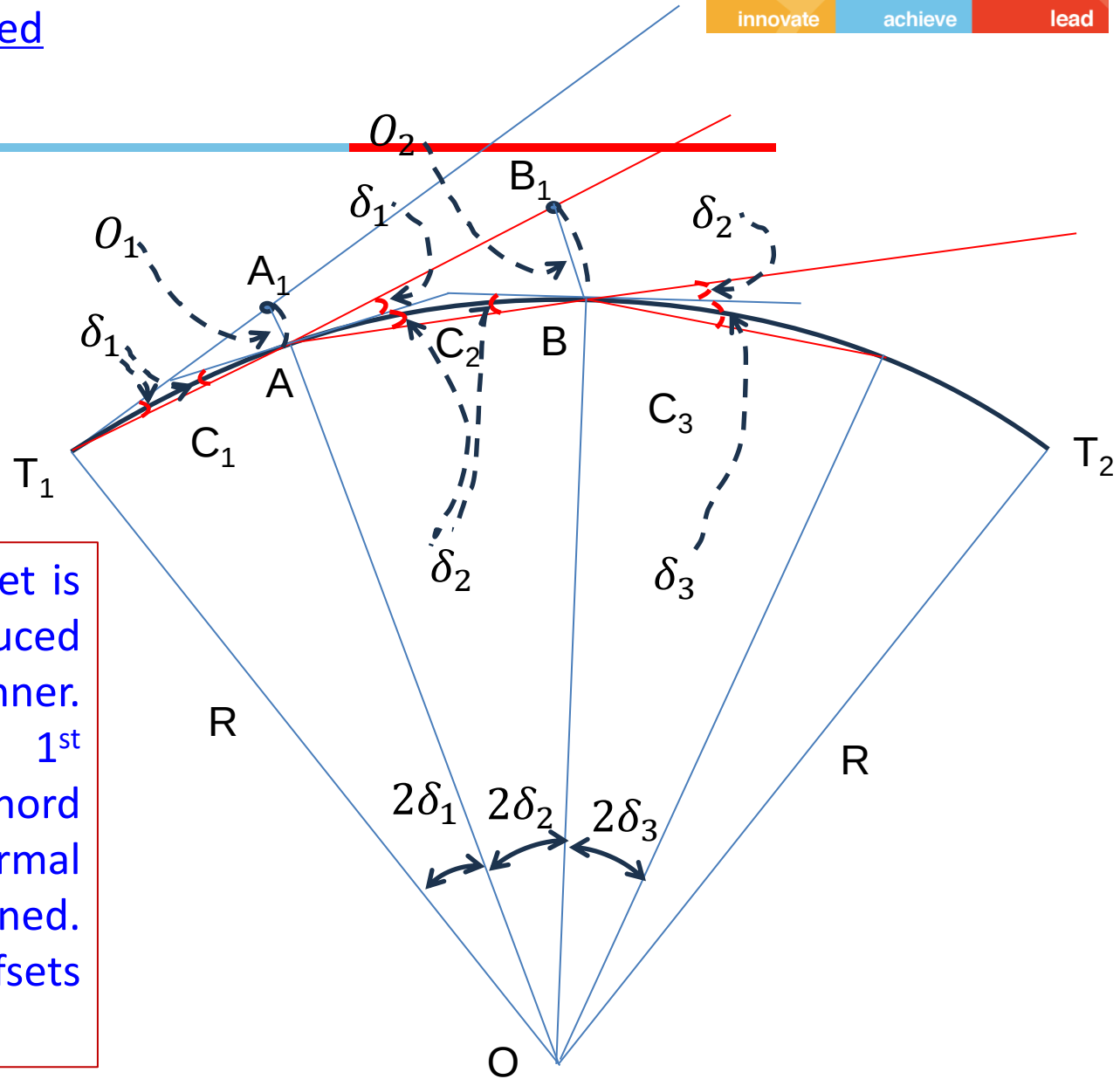
$$O_x = R\sqrt{1 + \left(\frac{x}{R}\right)^2} - R$$

$$O_x = R\left(1 + \frac{1}{2}\left(\frac{x^2}{R^2}\right)\right) - R$$

$$O_x = \frac{x^2}{2R}, \quad \text{Approximate Expression}$$

Successive bi-section of arcs or chords

Offset from Chord produced



In this method, the offset is taken from the produced chords in successive manner. First, the length of 1st sub-chord, last sub-chord and number of normal chords are determined. After that the offsets calculations are done.

C_1 - 1st sub chord

T_1A_1 is taken as C_1 and point A_1 is found.

Angle $T_1OA = 2\delta_1$

Arc $T_1A = R(2\delta_1) \cong$ Chord $T_1A = C_1$

$$\delta_1 = \frac{C_1}{2R}$$

Now, the 1st offset O_1 which is taken from the 1st tangent $\cong$ Arc $A_1A = C_1 \cdot \delta_1$

$$O_1 = C_1 \cdot \delta_1 = C_1 \cdot \frac{C_1}{2R} = \frac{C_1^2}{2R}$$

Now, the 2nd Offset O_2 is Arc B_1B which is taken from 1st produced chord,

$$O_2 = C_2 \cdot (\delta_1 + \delta_2) = C_2 \left(\frac{C_1}{2R} + \frac{C_2}{2R} \right) = \frac{C_2}{2R} (C_1 + C_2), \quad C_2 - 1^{\text{st}} \text{ normal chord.}$$

$$O_3 = C_3 \cdot (\delta_2 + \delta_3) = C_3 \left(\frac{C_2}{2R} + \frac{C_3}{2R} \right) = C_2 \left(\frac{C_2}{2R} + \frac{C_2}{2R} \right) = \frac{(C_2)^2}{R}, \quad \text{taken from 2nd produced chord}$$

C_1 - 1st sub chord

$C_2 = C_3 = C_4 = \dots = C_{n-1}$ = full or normal chords

C_n = last sub chord

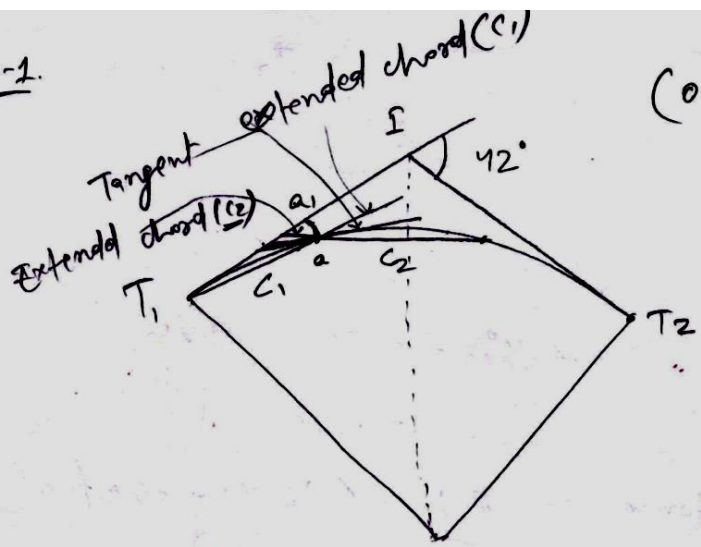
So, Offsets O_3 to O_{n-1} are of same value = $\frac{(C_2)^2}{R}$

$$O_n = \frac{C_n}{2R} (C_{n-1} + C_n)$$

Q1: Two straight roads intersect at a chainage of 3548.0m. The angle of deflection is 42° . These two roads are to be connected by a simple circular curve. The setting out work in the field will be done by offset from chord produced. For this work a 20m chain will be used. The length of the 1st sub chord is 17.5m. The length of the 1st offset produced from the tangent drawn at the point of curve is 1m. Find all other offsets required to set out the curve.



No-1.



(offset from chord produced method)

$C_1 = 17.5 \text{ m}$
 $O_1 = 1 \text{ m}$ } $O_1 = \frac{C_1^2}{2R}$ (3)

$\Rightarrow R = \frac{C_1^2}{2} \Rightarrow R = 153.125 \text{ m}$

Length of curve = $R \times \Delta$ (radian) $\frac{\pi}{180}$ (degree) $= 153.125 \times 42 \times \frac{\pi}{180}$
 $= 112.246 \text{ m}$ (3)

Length of 1st sub chord = 17.5 m, So the rest length is $112.246 - 17.5 = 94.746 \text{ m}$

So no of full chord = $\frac{94.746}{20} = 4.7373 \approx 4$ (4)

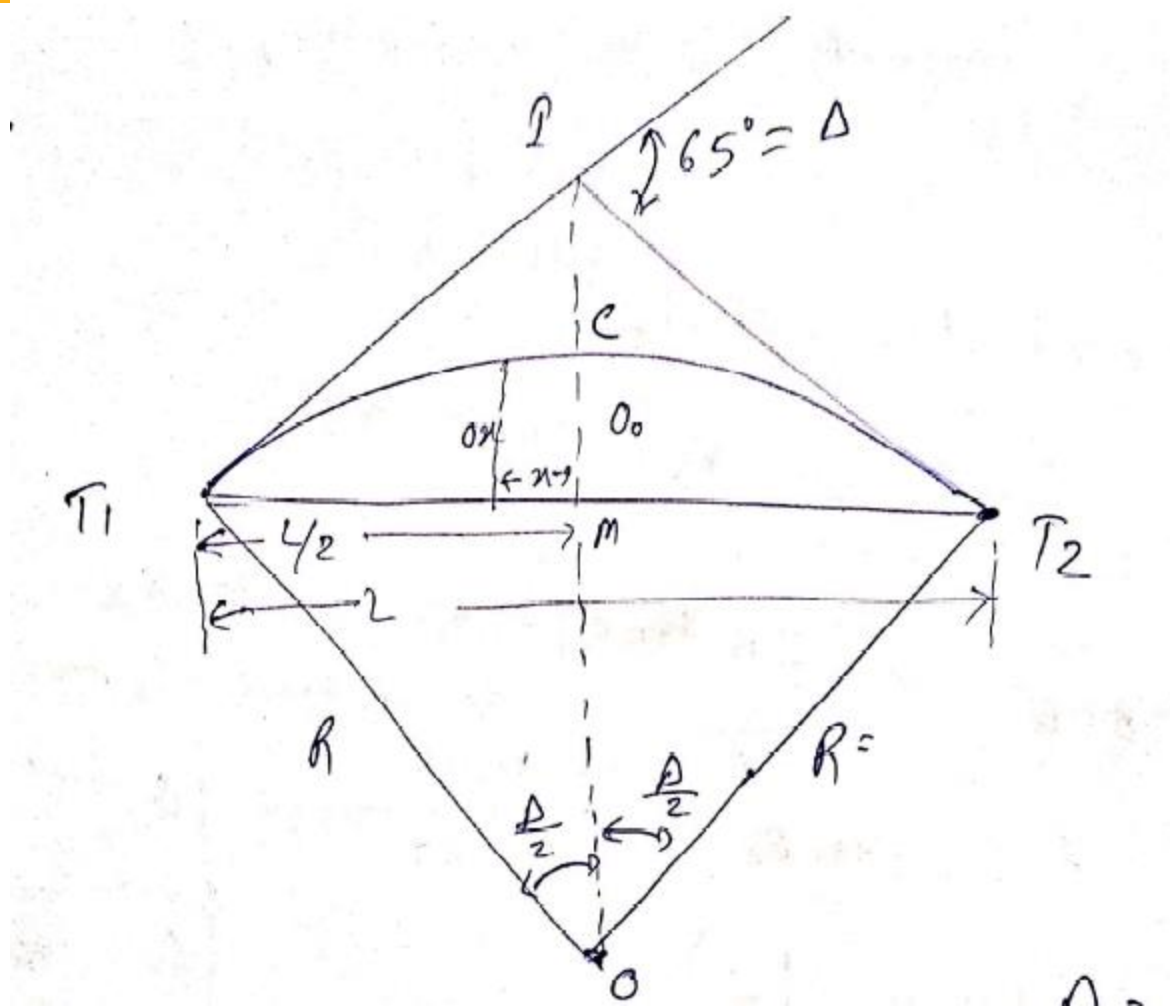
So, the last sub-chord = $0.7373 \times 20 = 14.746 \text{ m}$ (3)

Total chords = 6

$O_2 = \frac{C_2(C_2 + C_1)}{2R} = \frac{20(20 + 17.5)}{2 \times 153.125} = 2.4489 \text{ m}$ (2)
 $O_3 = O_4 = \dots = O_5 = \frac{C_3^2}{R} = 2.6122 \text{ m}$ (2)
 $O_6 = \frac{C_6(C_6 + C_5)}{2R} = 1.673 \text{ m}$ (2)

Q2: Two straight roads deflect at an angle of 65° . These two roads are to be connected with a 200m length simple circular curve. Find the necessary data to set out the curve by the following methods,

- (i) Offset from long chord with an interval of 10m (take both exact and approximate expressions)
- (ii) Offsets from chord produced (take chord length of 20m)



Length of curve $T_1CT_2 = 200\text{m}$, $\Delta = \text{Angle of deflection}$
 $\Delta = 65^\circ$

$R \Delta$ radian
 200m

$$R \times 65 \times \frac{\pi}{180} = 200$$

$$\Rightarrow \boxed{R = 176.294\text{m}} \quad (2)$$

$$\sin \frac{\Delta}{2} = \frac{T_1M}{R}$$

$$\Rightarrow T_1M = \left(\sin \frac{\Delta}{2} \right) R$$

$$\Rightarrow \left(\frac{L}{2} \right) = 99.722\text{m}$$

$$\boxed{L = 189.445\text{m} \leftarrow \text{Long chord.}} \quad (2)$$

(1) Offset from Long chord.

Mid ordinate $O_0 = R - \sqrt{R^2 - \left(\frac{L}{2} \right)^2} = 176.294$

$$= 176.294 - \sqrt{(176.294)^2 - (99.722)^2}$$

$$\boxed{O_0 = 27.608\text{m}} \quad (1)$$

Distance for offset (m)	O_x	Exact expression $O_x = \sqrt{R^2 - x^2} - (R - O_0)$	Approximate Expression $O_x = O_0 - \frac{x^2}{2R}$
0	O_0	27.608	27.608
10	O_{10}	27.324	27.324
20	O_{20}	26.469	26.473
30	O_{30}	25.036	25.055
40	O_{40}	23.010	23.070
50	O_{50}	20.368	20.517
60	O_{60}	17.083	17.397
70	O_{70}	13.115	13.710
80	O_{80}	8.411	9.456
90	O_{90}	2.904	4.635
94.722	$O_{94.722}$	-0.0007 $\approx$ 0.0	2.161

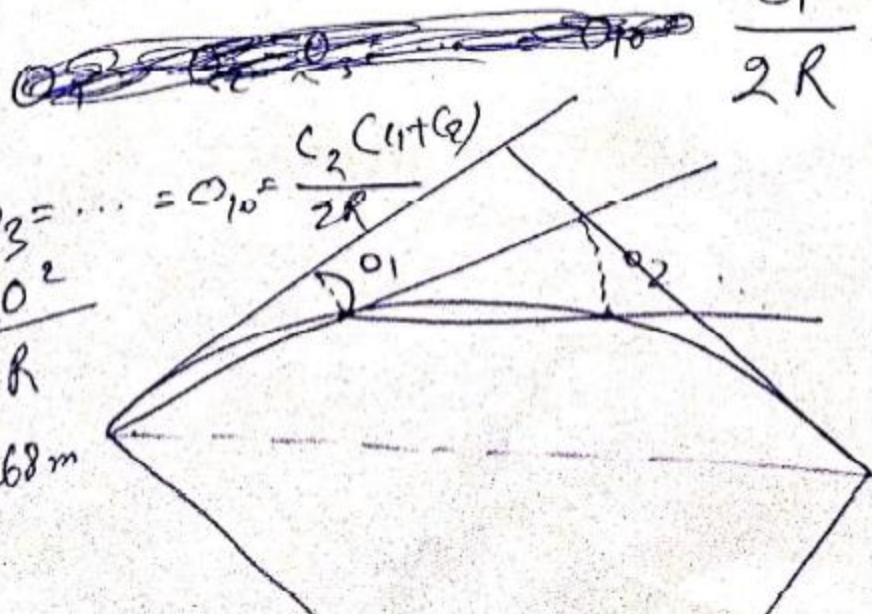
(ii) Offset from chord produced:

Curve length is 200m

And the chord length to be considered is 20m

So in total there are 10 offsets with equal chord length.

$$O_1 = \frac{C_1^2}{2R} = \frac{20^2}{2 \times 176.244} = 1.134 \text{ m}$$
$$O_2 = O_3 = \dots = O_{10} = \frac{C_2(C_1 + C_2)}{2R}$$
$$= \frac{20^2}{R} = 2.268 \text{ m}$$



5
0.5 mark
for each
offset.

Binomial Expansion

Let $\alpha = \frac{p}{q}$, where p and q are integers and $q \neq 0$. Then

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}x^3 + \dots$$

$\sqrt{1+x} = (1+x)^{\frac{1}{2}}$. Therefore, $\alpha = \frac{1}{2}$

$$\sqrt{1+x} = 1 + \frac{1}{2}x + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}x^2 + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{3!}x^3 + \dots$$

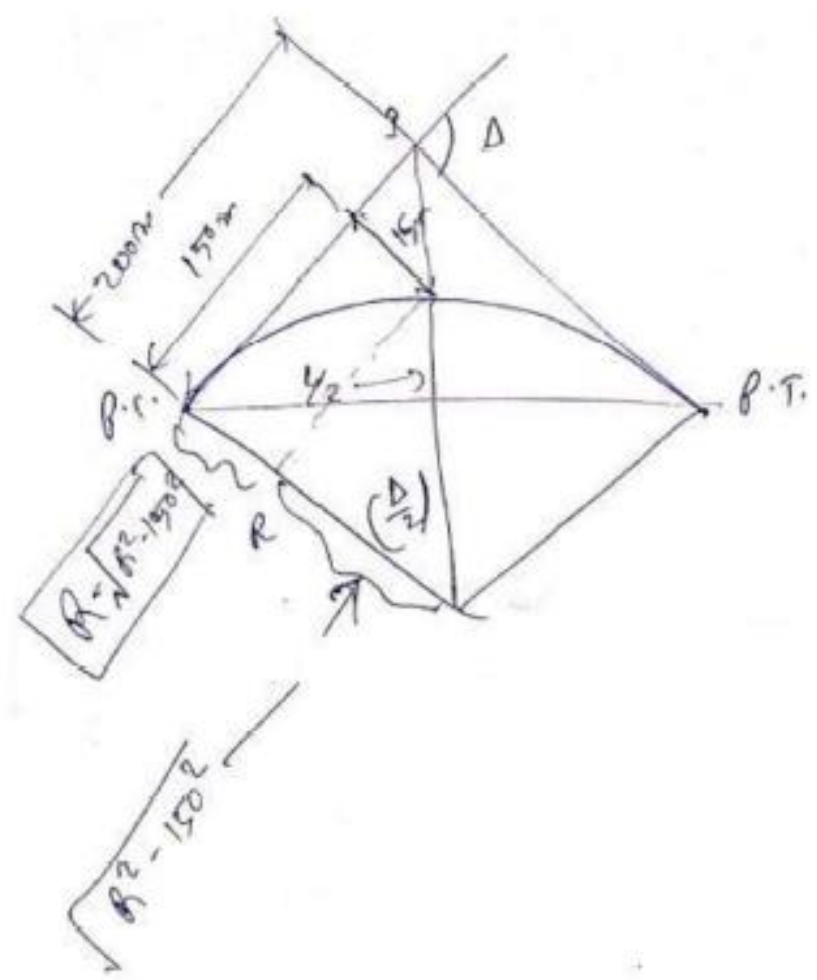
$$= 1 + \frac{1}{2}x + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2}x^2 + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{6}x^3 + \dots$$

$$= 1 + \frac{1}{2}x - \frac{\frac{1}{4}}{2}x^2 + \frac{\frac{3}{8}}{6}x^3 + \dots$$

$$= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + \dots$$

Q3: In setting out of a simple circular curve, the perpendicular offset from the tangent method is chosen. The tangent length is 200m. The perpendicular offset at a distance of 150m from the point of curve along the tangent length is 15m. Find the other eight perpendicular offsets starting from the point of curve at a distance of every 15m along the tangent. Find the length of the curve. Use exact expressions and approximate expressions.

Solution



$$R - \sqrt{R^2 - 150^2} = 15$$

Implies,

$$R - 15 = \sqrt{R^2 - 150^2}$$

$$\Rightarrow (R - 15)^2 = R^2 - 150^2$$

$$\Rightarrow R = 757.5m$$

Now

$$\tan \frac{\Delta}{2} = \frac{200}{757.5}$$

$$\Rightarrow \Delta = 29.58 \text{ degrees}$$

Therefore,

$$\sin \frac{\Delta}{2} = \frac{L/2}{R}$$

$$\Rightarrow \frac{L}{2} = R \sin \frac{\Delta}{2} = 193.372$$

$$\Rightarrow \text{Length of long chord} = L = 386.744m$$

$$\Rightarrow \text{Length of Curve} = R \Delta = 391.073m$$

x	Offset ($R - \sqrt{R^2 - 150^2}$)
0	0
15	0.148
30	0.594
45	1.338
60	2.379
75	3.722
90	5.365
105	7.312

Q4: Two straight roads meet at an angle of 130° . Calculate the necessary data for setting out a curve of 15 chain radius to connect these two straight roads. It is intended to follow the method of offset from chord. The chain length is 20m. Find the offsets by exact expression and approximate expression.

Solution:

Q5: Determine the ordinates of the points on a circular curve having long chord of 100m and mid-ordinate of 5m. The ordinates are to be measured from long chord at an interval of 10m. Use both exact expression and approximate expression.

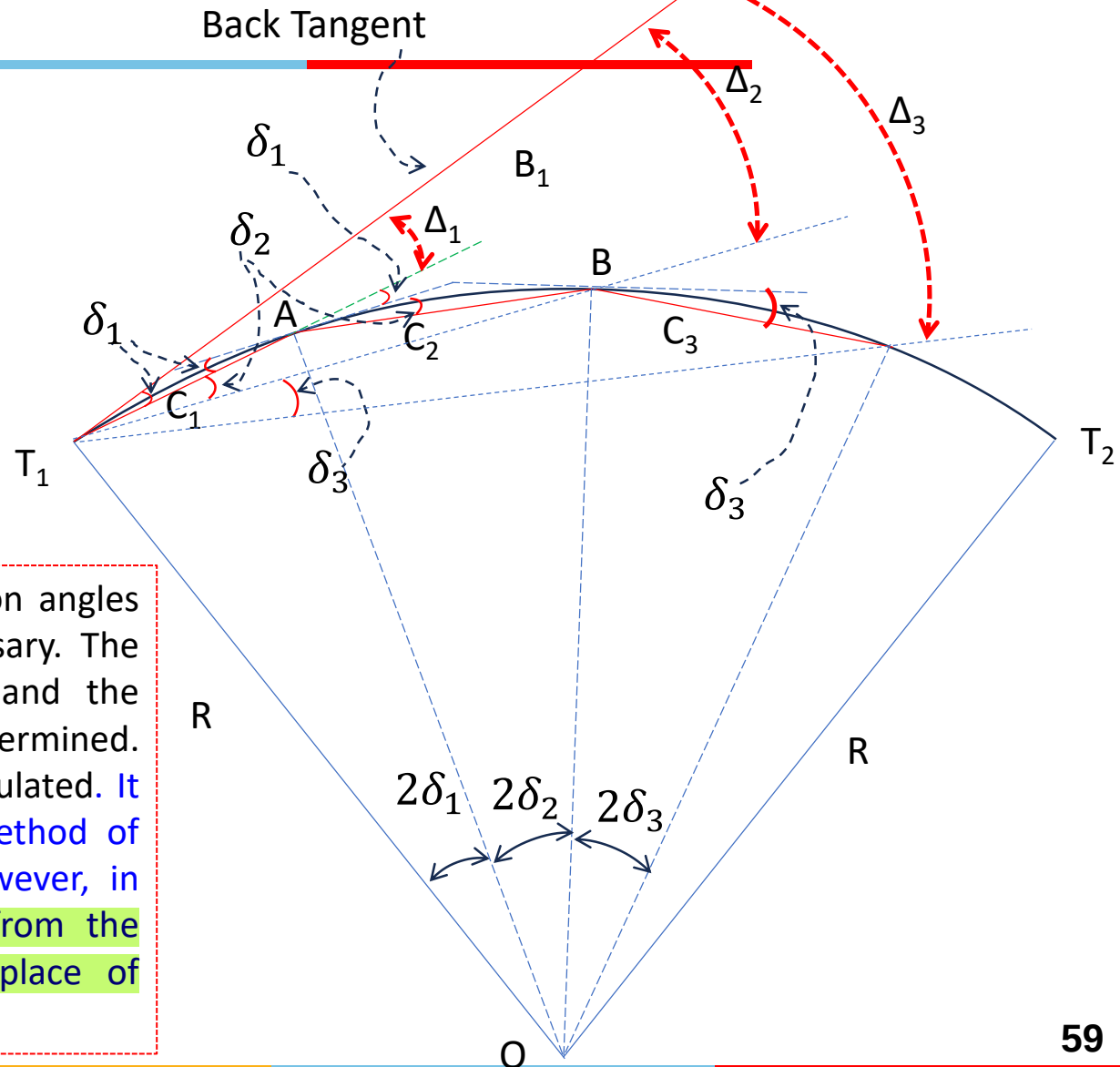
Solution:

Angular Methods

The various angular methods are,

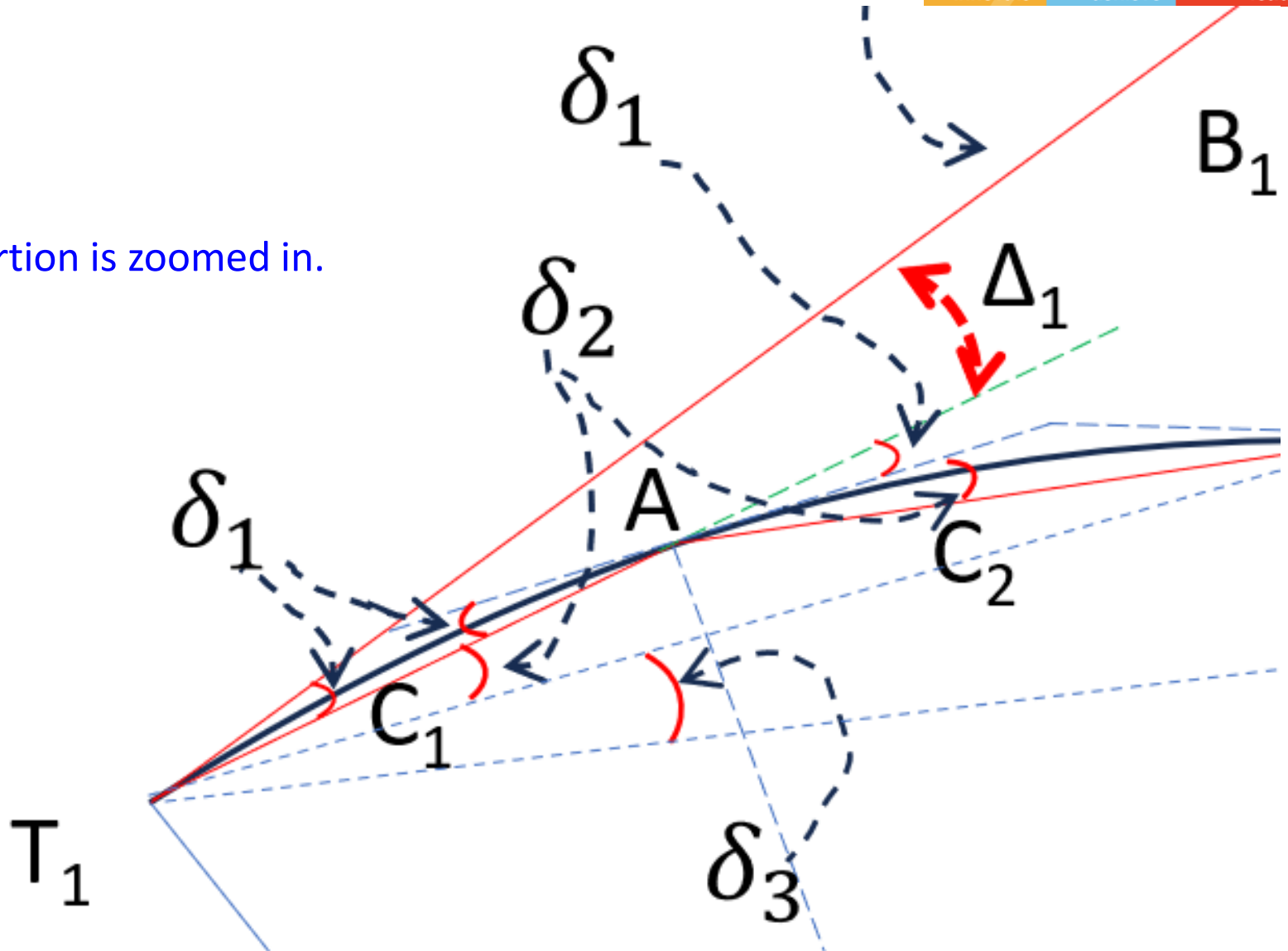
- (i) Rankine's Method of tangential(deflection) angle {One theodolite method}
- (ii) Two theodolite method
- (iii) Tacheometry method

Rankine's Method of tangential(deflection) angle {One theodolite method}



In this method, both the deflection angles and the chord lengths are necessary. The first sub-chord, last sub-chords and the number of normal chords are determined. Then the deflection angle are calculated. It is more or less similar to the method of offset from chord produced. However, in this case the deflection angles from the back tangent are calculated in place of offsets from the produced chords.

Left portion is zoomed in.



C_1 - 1st sub chord

Angle $T_1OA = 2\delta_1$

Arc $T_1A = R(2\delta_1) \cong$ Chord $T_1A = C_1$

$\delta_1 = \frac{C_1}{2R}$ radian

Similarly, Arc $AB = R(2\delta_2) \cong$ Chord $AB = C_2$

$\delta_2 = \frac{C_2}{2R}$ radian

C_1 - 1st sub chord

Angle $T_1OA = 2\delta_1$

Arc $T_1A = R(2\delta_1) \cong$ Chord $T_1A = C_1$

$$\delta_1 = \frac{C_1}{2R} \text{ radian}$$

Similarly, Arc $AB = R(2\delta_2) \cong$ Chord $AB = C_2$

$$\delta_2 = \frac{C_2}{2R} \text{ radian}$$

All subtended angles are calculated with respect to the first sub-chord, with respect to the all normal chords and with respect to the last sub-chord. Then the deflection angles are calculated adding the subtended angles gradually.

The first subtended angle and the last subtended angle are different because those are with respect to the first sub-chord and the last sub-chord. All the intermediate subtended angles are same because those are calculated with respect to the normal chords.

The deflection angles are calculated as,

$$\Delta_1 = \delta_1$$

$$\Delta_2 = \delta_1 + \delta_2 = \Delta_1 + \delta_2$$

$$\Delta_3 = \delta_1 + \delta_2 + \delta_3 = \Delta_2 + \delta_3$$

$$\Delta_4 = \delta_1 + \delta_2 + \delta_3 + \delta_4 = \Delta_3 + \delta_4$$

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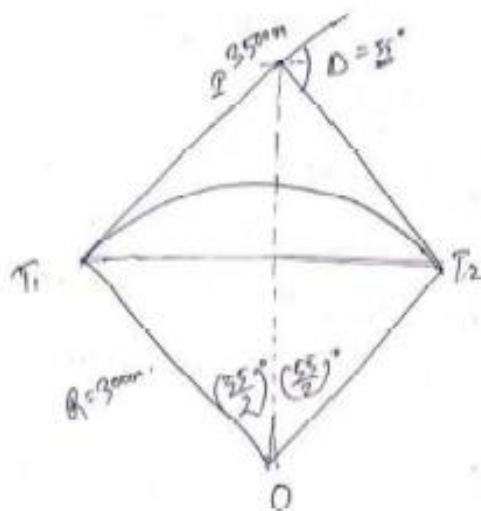
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$$\Delta_n = \Delta_{n-1} + \delta_n = \frac{\Delta}{2}, \text{ it can be a check.}$$

Q6. The deflection angle of two straight roads is 55° and these two roads meet at a chainage of 3500m. To join both the roads by a simple circular curve with radius 300m, a setting out work has to be carried out. Find the required data for this work by Rankins's method of deflection angle. Consider chord interval of 30m.

Solution



$$\text{Length} = R \times \Delta \quad (\Delta \text{ in radians})$$

$$= 300 \times 55 \times \frac{\pi}{180} = 287.979 \text{ m}$$

$$\tan \frac{\Delta}{2} = \frac{IT_1}{R}$$

$$IT_1 = \text{tangent length} = \tan \frac{\Delta}{2} R = 156.170 \text{ m}$$

$$\text{Chainage at } T_1 = 3500 - 156.170 \text{ m}$$

$$= 3343.83 \text{ m}$$

$$= 111.461 \text{ chains}$$

So 1st chord will be a length for next chain full length.

$$\text{Means } (1 - 0.461) = 0.539 \text{ chain}$$

$$\text{Means } C_1 = 16.17 \text{ m (1st sub chord)}$$

$$C_2 = C_3 = C_4 = \dots = C_{10} = 30 \text{ m (a full-chord)}$$

$$C_{11} = 1.809 \text{ m (last sub chord)}$$

$$\delta_1 = \frac{C_1}{2R} \text{ radian} = \frac{16.17}{2 \times 300} = 0.026 \text{ radian}$$

$$\delta_2 = \delta_3 = \dots = \delta_{10} = \frac{30}{2 \times 300} = 0.05 \text{ radian}$$

$$\delta_{11} = \frac{1.809}{2 \times 300} = 0.003 \text{ radian}$$

$$\Delta_1 = \delta_1 = 0.026 \text{ radian} = 1.489 \text{ degree} = 1 \text{ degree } 29 \text{ minutes } 20.4 \text{ seconds}$$

$$\Delta_2 = \Delta_1 + \delta_2 = 0.026 + 0.05 = 0.075 \text{ radian} = 4 \text{ deg } 17 \text{ min } 49.86 \text{ sec}$$

$$\Delta_3 = \Delta_2 + \delta_3 = 0.075 + 0.05 = 0.125 \text{ radian} = 7 \text{ deg } 9 \text{ min } 43.1 \text{ sec}$$

$$\Delta_4 = \Delta_3 + \delta_4 = 0.125 + 0.05 = 0.175 \text{ radian} = 10 \text{ deg } 1 \text{ min } 36.341 \text{ sec}$$

$$\Delta_5 = \Delta_4 + \delta_5 = 0.175 + 0.05 = 0.225 \text{ radian}$$

$$\Delta_6 = \Delta_5 + \delta_6 = 0.225 + 0.05 = 0.275 \text{ radian}$$

$$\Delta_7 = \Delta_6 + \delta_7 = 0.275 + 0.05 = 0.325 \text{ radian}$$

$$\Delta_8 = \Delta_7 + \delta_8 = 0.325 + 0.05 = 0.375 \text{ radian}$$

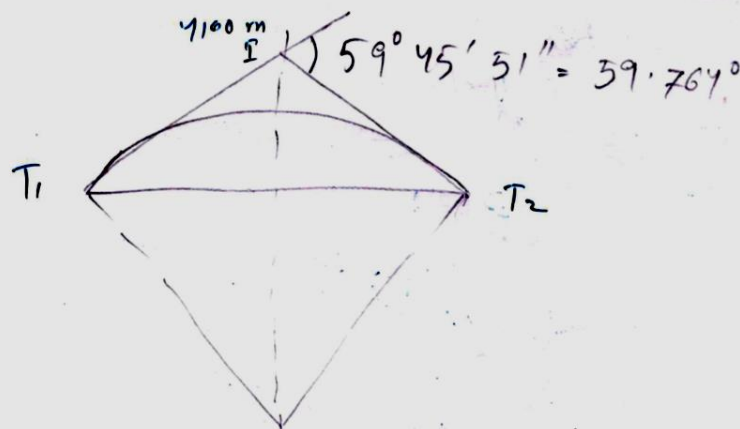
$$\Delta_9 = \Delta_8 + \delta_9 = 0.375 + 0.05 = 0.425 \text{ radian}$$

$$\Delta_{10} = \Delta_9 + \delta_{10} = 0.425 + 0.05 = 0.475 \text{ radian}$$

$$\Delta_{11} = \Delta_{10} + \delta_{11} = 0.475 + 0.003 = 0.478 \text{ radian}$$

Q7. The deflection angle of two straight roads is $59^{\circ} 45' 51''$ and these two roads meet at a chainage of 4100m. To join both the roads a simple circular curve is chosen, whose degree of curve is 4. A setting out work has to be carried out. Find the required data for this work by Rankin's method of deflection angle. Consider chord interval of 30m.

Q10-2



Degree of curve is 4° for 30 m chain,

$$R = \frac{1719}{\Delta} = \frac{1719}{4} = 429.75 \text{ m} \quad (2)$$

$$\text{Tangent distance } T_1 = R \cdot \tan\left(\frac{\Delta}{2}\right) = 429.75 \times \tan\left(\frac{39.764}{2}\right) = 246.847 \text{ m} \quad (1)$$

$$\text{Chainage of 1st tangent point is } 4100 - 246.847 = 3853.153 \text{ m} \quad (1)$$

Length of 1st sub chord:

$$\text{No. of chain at } T_1 = \frac{3853.153}{30} = 128.438 \text{ chains. So the next point will be at } 129 \text{ chains.} \quad (2)$$

$$\text{So 1st sub chord} = (129 - 128.438) \times 30 = 16.86 \text{ m} \quad (2)$$

$$\text{Length of Curve} = R \cdot \Delta = R \times \Delta \times \frac{\pi}{180}$$

$$= 429.75 \times 39.764 \times \frac{\pi}{180}$$

$$= 448.128 \text{ m} \quad (2)$$

$$\text{No. of full chord} = \frac{448.128 - 16.86}{30} = 14.375 \approx 14 \quad (14)$$

innovate

achieve

lead

Last sub-chord = $0.375 \times 30 = 11.25 \text{ m}$

Total chords = 16

The deflection angles.

$$\delta_1 = \frac{C_1}{2R} = \frac{16.86}{2 \times 429.75} = 0.0196 \text{ radian} = 1.1229 \text{ degree} = \Delta_1$$

$$\delta_2 = \frac{C_2}{2R} = \frac{30}{2 \times 429.75} = 1.999 \text{ degree}$$

$$\Delta_2 = \Delta_1 + \delta_2 = 1.1229 + 1.999 = 3.1219 \text{ degree}$$

$$\Delta_3 = \Delta_2 + \delta_3 = 3.1219 + 1.999 = 5.1209 \text{ degree}$$

$$\Delta_4 = \Delta_3 + \delta_4 = 5.1209 + 1.999 = 7.1199 \text{ degree}$$

$$\Delta_5 = \Delta_4 + \delta_5 = 7.1199 + 1.999 = 9.1189 \text{ degree}$$

$$\Delta_6 = \Delta_5 + \delta_6 = 9.1189 + 1.999 = 11.1179 \text{ degree}$$

$$\Delta_7 = \Delta_6 + \delta_7 = 11.1179 + 1.999 = 13.1169 \text{ degree}$$

$$\Delta_8 = \Delta_7 + \delta_8 = 13.1169 + 1.999 = 15.1159 \text{ degree}$$

$$\Delta_9 = \Delta_8 + \delta_9 = 15.1159 + 1.999 = 17.1149 \text{ degree}$$

$$\Delta_{10} = \Delta_9 + \delta_{10} = 17.1149 + 1.999 = 19.1139 \text{ degree}$$

$$\Delta_{11} = \Delta_{10} + \delta_{11} = 19.1139 + 1.999 = 21.1129 \text{ degree}$$

$$\Delta_{12} = \Delta_{11} + \delta_{12} = 21.1129 + 1.999 = 23.1119 \text{ degree}$$

$$\Delta_{13} = \Delta_{12} + \delta_{13} = 23.1119 + 1.999 = 25.1109 \text{ degree}$$

$$\Delta_{14} = \Delta_{13} + \delta_{14} = 25.1109 + 1.999 = 27.1089 \text{ degree}$$

$$\Delta_{15} = \Delta_{14} + \delta_{15} = 27.1089 + 1.999 = 29.1089 \text{ degree}$$

$$\delta_{16} = \frac{C_{16}}{2R} = \frac{11.25}{2 \times 429.75} = 0.7499 \text{ degree}$$

$$\Delta_{16} = \Delta_{15} + \delta_{16} = 29.1089 + 0.7499 = 29.8588 \text{ degree} = \left(\frac{A}{2}\right)$$



Two theodolite method

Transition curve

In simple circular curve, sudden change from straight path to circular path becomes necessary. However, it is not possible to provide this change instantaneously. So, from the straight path to the circular path a curve with gradual change in radius is provided. This curve with varying radius is known as transition curve. The radius of such curve varies from infinity to certain fixed value. This curve is called easement curve also.

If the transition curve is not provided then the passengers experience a shock when the vehicle passes from the straight road to the circular road. At the junction of straight road and circular road the curvature suddenly changes from zero to some definite value. The centrifugal force comes into picture. If the centrifugal force exceeds some value the vehicle may overturn.

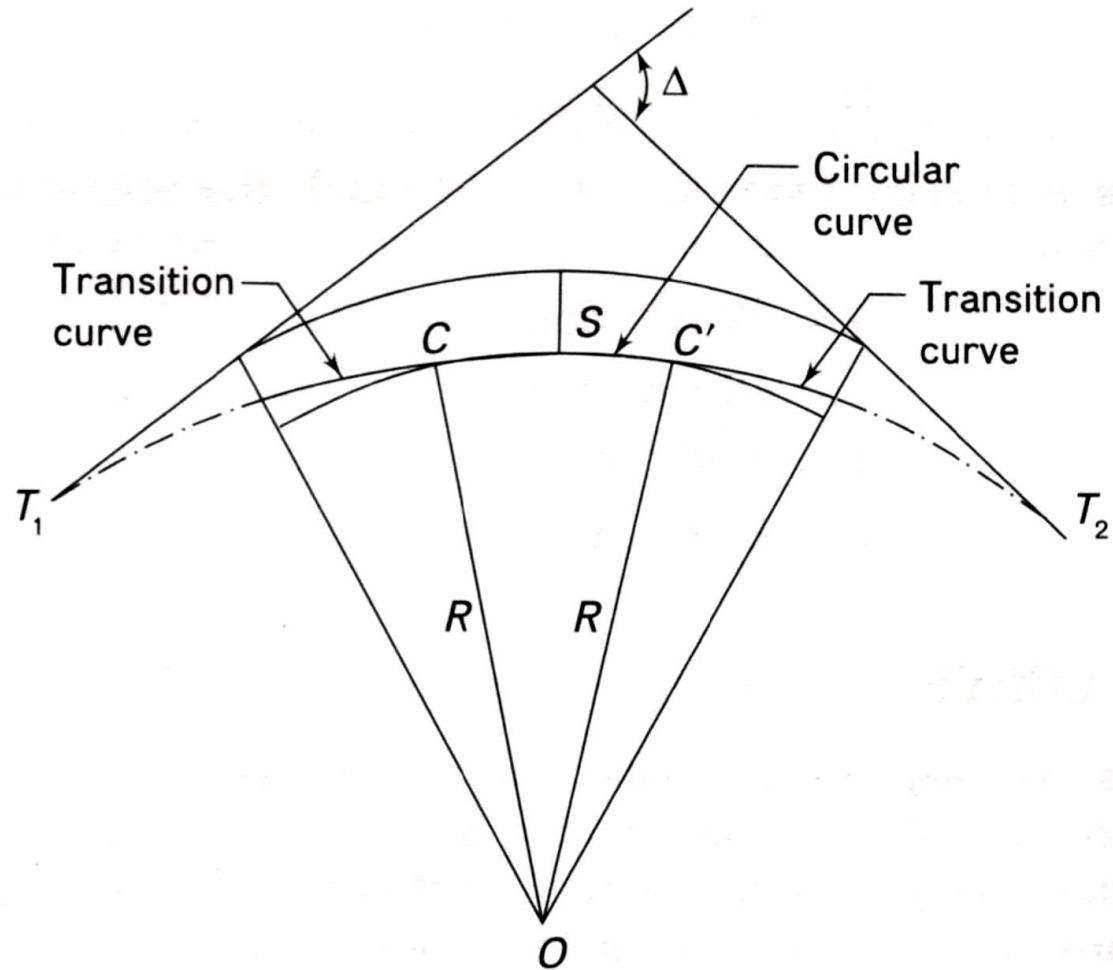
In railways the outer rail will take the shock and will wear at the junction point.

To overcome all these difficulties the transition curve is provided.

Why there will be a shock when the vehicle traverses from straight road to circular road ?

Because, immediately the superelevation will start at the junction.

$$curvature = \frac{1}{radius\ of\ curvature}$$



Objective(Purpose) of providing transition curves

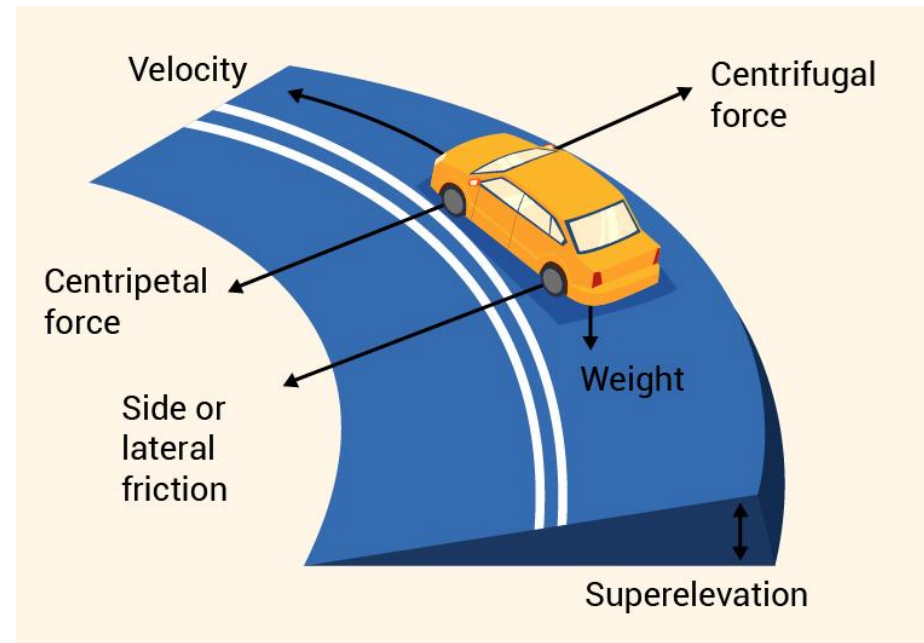
1. To achieve gradual transition from the straight path to the circular path and from the circular path to the straight path.
2. To obtain a gradual increase of curvature from zero at the tangent point to the specified quantity at the junction of the transition curve with the circular curve.
3. To provide the superelevation gradually from zero at the tangent point to the specified amount on the circular curve.
4. To avoid the overturning of the vehicle.
5. To provide extra widening on the circular path.

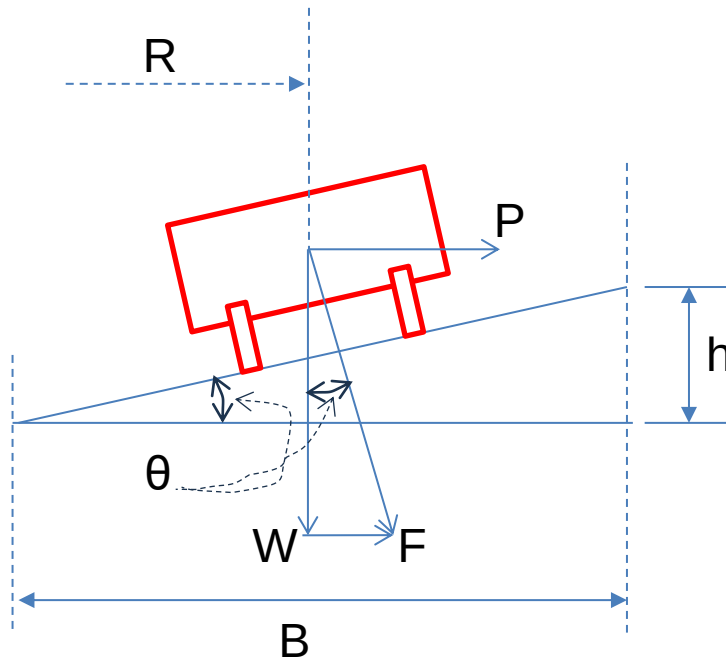
Requirement of ideal transition curves

1. Should originate tangentially from the straight paths. (from points T_1 and T_2)
2. Should meet the circular curve tangentially. (at C and C' points)
3. The radius of the transition curve is infinity at the origin i.e. at points T_1 and T_2 .
4. The radius of the transition curve at C and C' points is same as that of the radius of the circular curve.
5. The length of the transition curve should be such the required superelevation is attained at C and C'.
6. The rate of increase of curvature along the transition curve should be same as the rate of increase of superelevation.

Superelevation

The superelevation is defined as the raising of the outer edge of a road or railway over the inner edge.





Friction is neglected(frictionless case)

$$P = ma_r = m \frac{v^2}{R} = \frac{Wv^2}{gR}$$

$$\frac{P}{W} = \frac{v^2}{gR}$$

$$\tan \theta = \frac{h}{B}$$

$$\tan \theta = \frac{P}{W} = \frac{v^2}{gR}$$

$$\frac{h}{B} = \frac{v^2}{gR}$$

$$v = \sqrt{gR \tan \theta}$$

$$h = \frac{Bv^2}{gR}$$

There is a limit to maximum speed and minimum speed without skidding, considering friction into account. In the superelevated road, any minimum speed may not be safe.

Can a vehicle be kept standing(zero velocity) on a superelevated road ?

Reference (maximum velocity and minimum velocity when coefficient of friction is considered)

$$v_{max} = \sqrt{\frac{gR(\tan\theta + \mu_s)}{(1 - \mu_s)}}$$

$$v_{min} = \sqrt{\frac{gR(\tan\theta - \mu_s)}{(1 + \mu_s)}}$$

Coefficient of friction between tyer and road surface is μ_s

Our case is without friction

Where,

P- Centrifugal force

W- Weight of the body (mg)

B - Width of the road

h – Superelevation

R- Radius of the curve

θ – Angle of superelevation

g – Acceleration due to gravity

m – mass of the body

R – Resultant

v – speed of the vehicle

Length of transition curve

The different method used to calculate the length of transition curve are,

- (i) By the rate of superelevation
- (ii) By time rate
- (iii) By the rate of change of radial acceleration
- (iv) Centrifugal ratio and comfort condition

By the rate of superelevation

For 1 cm of superelevation n meter of transition curve length provided

So, for h cm of superelevation $n \times h$ meter of transition curve length provided

Length of transition curve = $L = n \times h$

(How to decide n ??) (where, $h = \frac{Bv^2}{gR}$)

By time rate

x cm of superelevation is applied in 1 sec

So, for h cm of superelevation is applied in $\frac{h}{x}$ sec (say, $\frac{h}{x} = t$ sec)

$$\text{Length of transition curve} = L = \text{distance travelled} = vt = v \frac{h}{x} = v \frac{Bv^2}{xgR} = \frac{Bv^3}{xgR}$$

(How to decide x ??) (where, $h = \frac{Bv^2}{gR}$)

By the rate of change of radial acceleration

The rate of change of radial acceleration should be small, constant, and consistent.

The length of the transition curve is calculated for a specified rate of change of radial acceleration. That means the rate of change of radial acceleration is kept constant through the curve.

Also, the rate of change of radial acceleration should be such that the passenger should not sense any sign of discomfort, while travelling on the curve.

rate of change of radial acceleration is α

radial acceleration is $\frac{v^2}{R}$

Now, rate of change of radial acceleration is $\frac{d\left(\frac{v^2}{R}\right)}{dt} = \alpha$, and α is constant throughout the curve

So, we can say $\frac{\frac{v^2}{R}}{t} = \alpha$, $\alpha = \frac{v^2}{tR}$, $t = \frac{v^2}{\alpha R}$

If length of the transition curve is L , the speed of the vehicle is v and time required to travel this length is t . Then $L = vt$

$$L = vt = v \frac{v^2}{\alpha R} = \frac{v^3}{\alpha R}, \quad \text{as } t = \frac{v^2}{\alpha R}$$

(α) is normally kept 0.3 for comfort condition.

Length of transition curve(L) varies inversely with the radius of the circular curve(R) if the velocity is independent of radius(R).

In sharp curves, the angle of superelevation(θ) increases. It becomes an issue for the vehicle standing on these curves. So the superelevation is reduced and the maximum speed of the vehicle is reduced on these curves. This reduction in superelevation also helps vehicle to stand safely.

Centrifugal ratio and comfort condition

In Roads, for comfort condition, the centrifugal ($\frac{P}{W}$) is kept $\frac{1}{4}$

$$\tan \theta = \frac{P}{W} = \frac{v^2}{gR} = \frac{1}{4} \quad \Rightarrow \quad v = \sqrt{\frac{gR}{4}}$$

$$\text{Length of Transition curve} = L = \frac{v^3}{\alpha R}, \quad L = \frac{\left(\sqrt{\frac{gR}{4}}\right)^3}{\alpha R} = \frac{(\sqrt{gR})^3}{(\sqrt{4})^3 \alpha R} = \frac{(\sqrt{g})^3 (\sqrt{R})^3}{(2)^3 \alpha R} = \frac{(g)^{1.5} \sqrt{R}}{(2)^3 \alpha}$$

Taking, $\alpha = 0.3$ for comfort condition and $g=9.81\text{m/sec}^2$, $L = 12.802 \sqrt{R}$, R in m

In Railways, for comfort condition, the centrifugal ($\frac{P}{W}$) is kept $\frac{1}{8}$

$$\text{So, Length of Transition curve} = L = \frac{\left(\sqrt{\frac{gR}{8}}\right)^3}{\alpha R} = \frac{(\sqrt{gR})^3}{(\sqrt{8})^3 \alpha R} = \frac{(\sqrt{g})^3 (\sqrt{R})^3}{(2\sqrt{2})^3 \alpha R} = \frac{(g)^{1.5} \sqrt{R}}{(2\sqrt{2})^3 \alpha}$$

Taking, $\alpha = 0.3$ for comfort condition and $g=9.81\text{m/sec}^2$, $L = 4.5 \sqrt{R}$, R in m

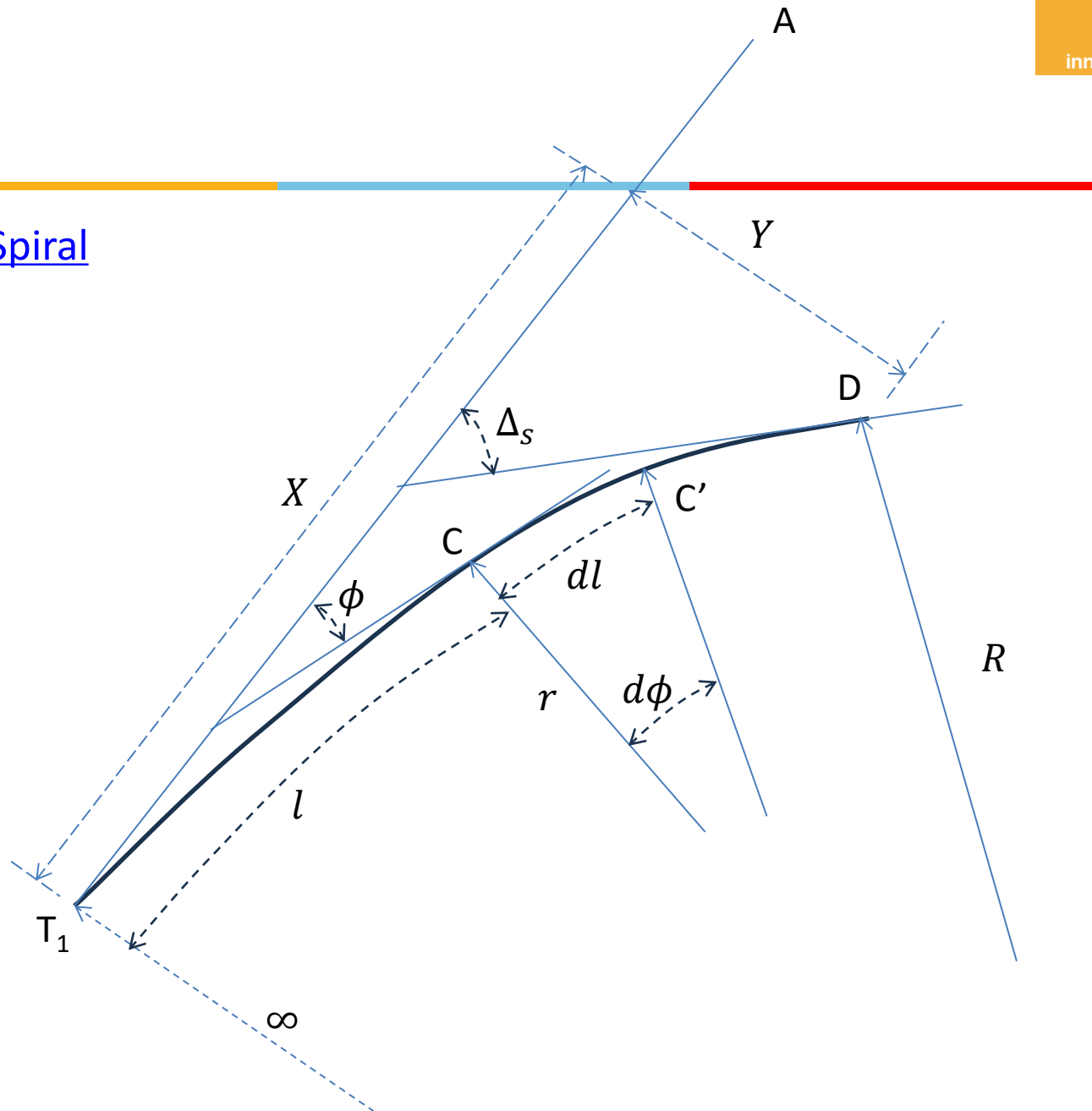
Types of Transition Curves

The different type of transition curves are,

- (i) Cubic Spiral or Clothoid or Euler Spiral
- (ii) Bernoulli's lemniscate
- (iii) Cubic parabola or Froud's curve

All there give more or less the same curve on the ground. However, the cubic spiral is the ideal transition curve. The rate of change of radial acceleration remains constant.

Cubic Spiral



Where,

T_1 - beginning of the transition curve

$T_1 A$ – 1st tangent (or initial tangent)

D – junction of transition curve and the circular curve

C - any point on the curve at (l) along the curve from the beginning of the transition curve(T_1) and it has a radius (r).

C' – neighbourhood point of C

(ϕ) – angle between 1st tangent and tangent at C

(Δ_s) – angle between 1st tangent and the tangent at the junction point D.

R- radius of the transition curve at the end of the curve

The radius of the curve is inversely proportional to the distance of the curve from the beginning.

$$l \propto \frac{1}{r}$$

$$lr = \text{constant} = LR$$

From the curve we can take,

$$d\phi \cdot r = dl$$

$$d\phi = \frac{1}{r} dl \quad (lr = \text{constant} = LR, \text{ so } \frac{1}{r} = \frac{l}{LR})$$

$$d\phi = \frac{l}{LR} dl$$

Integrating both sides, we get

$$\phi = \frac{1}{LR} \left[\frac{l^2}{2} \right] + C_1 \quad \{ C_1 - \text{is constant of integration} \}$$

at $l = 0, \phi = 0,$ so $C_1 = 0$

$$\phi = \frac{l^2}{2LR} \quad \text{also,} \quad l^2 = 2RL\phi \quad \text{and} \quad l = \sqrt{2RL\phi}$$

at $l = L, \phi = \Delta_s$

$$\Delta_s = \frac{L}{2R}$$



$$\cos \phi = \frac{dx}{dl}$$

$$dx = dl \cos \phi$$

$$dx = dl \cos \phi$$

$$dx = dl \left(1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} - \dots \right)$$

$$\text{we know, } l = \sqrt{2RL\phi} = \sqrt{2RL} \cdot \sqrt{\phi}$$

$$\text{So, } \frac{dl}{d\phi} = \sqrt{2RL} \frac{d}{d\phi} \sqrt{\phi} = \sqrt{2RL} \frac{d}{d\phi} \phi^{\frac{1}{2}} = \sqrt{2RL} \frac{1}{2} \phi^{\frac{1}{2}-1} = \frac{\sqrt{2RL}}{2\sqrt{\phi}}$$

$$dl = \frac{\sqrt{2RL}}{2\sqrt{\phi}} d\phi$$

$$dx = \frac{\sqrt{2RL}}{2\sqrt{\phi}} d\phi \left(1 - \frac{\phi^2}{2!} + \frac{\phi^4}{4!} - \dots \right)$$

$$dx = \frac{\sqrt{2RL}}{2} \left(\frac{1}{\sqrt{\phi}} - \frac{\phi^2}{2\sqrt{\phi}} \right) d\phi, \text{ neglecting the higher order terms}$$

$$dx = \frac{\sqrt{2RL}}{2} \left(\phi^{-\frac{1}{2}} - \frac{\phi^{2-\frac{1}{2}}}{2} \right) d\phi$$

$$dx = \frac{\sqrt{2RL}}{2} \left(\phi^{-\frac{1}{2}} - \frac{\phi^{\frac{3}{2}}}{2} \right) d\phi$$

Integrating, We shall get,

$$x = \frac{\sqrt{2RL}}{2} \left(\frac{\phi^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} - \frac{1}{2} \cdot \frac{\phi^{\frac{3}{2}+1}}{\frac{3}{2}+1} \right)$$

$$x = \frac{\sqrt{2RL}}{2} \left(\frac{\phi^{\frac{1}{2}}}{\frac{1}{2}} - \frac{1}{2} \cdot \frac{\phi^{2.5}}{2.5} \right)$$

$$x = \sqrt{2RL} \left(\frac{\phi^{\frac{1}{2}}}{1} - \frac{\phi^{2.5}}{10} \right)$$

$$x = \sqrt{2RL} \sqrt{\phi} \left(1 - \frac{\phi^2}{10} \right)$$

$$x = l \left(1 - \frac{\phi^2}{10} \right) \text{-----(1)} \quad \text{as } \sqrt{2RL\phi} = l$$

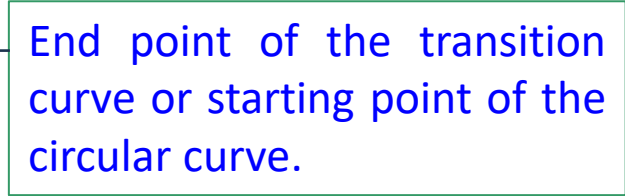
Similarly, we can take,

$\sin \phi = \frac{dy}{dl}$ and performing all the steps we get ,

$$y = \frac{l^3}{6RL} \left(1 - \frac{\phi^2}{14} \right) \text{-----}(2)$$

From Eqⁿ (1) and From Eqⁿ (2) , if we take the 1st terms only, we get,

$$x = l \text{ and } y = \frac{x^3}{6RL} \text{ which is Equation of cubic parabola}$$



C is any point on the transition curve at x and y . C' is the neighbourhood point.

(ϕ) – Angle made by the tangent at C and the first tangent at T_1 . It is called **spiral angle**.

(α) – Angle made by the chord T_1C and the first tangent at T_1 . It is called **deflection angle**.

l – Curved length of the transition curve up to point C.

We know, $y = \frac{x^3}{6RL}$

$$\tan \alpha = \frac{y}{x} = \frac{x^3}{6RL x} = \frac{x^2}{6RL} \text{ -----(i)}$$

$$\tan \phi = \frac{dy}{dx}$$

$$\tan \phi = \frac{dy}{dx} = \frac{d\left(\frac{x^3}{6RL}\right)}{dx} = \frac{x^2}{2RL} \text{ -----(ii)}$$

From, Eqn (i) and Eqn (ii), we can say

$$\tan \alpha = \frac{1}{3} \tan \phi$$

$$\alpha = \frac{1}{3} \phi, \text{ as both } \alpha \text{ and } \phi \text{ both are small.}$$

We know , $\sqrt{2RL\phi} = l$ or $l^2 = 2RL\phi \Rightarrow \phi = \frac{l^2}{2RL}$

$$\alpha = \frac{l^2}{6RL}$$

At, point D, End point of the transition curve or starting point of the circular curve,

l becomes L , α becomes α_s and ϕ becomes Δ_s

$$\alpha_s = \frac{L^2}{6RL} = \frac{L}{6R} \quad \text{and} \quad \Delta_s = \frac{L^2}{2RL} = \frac{L}{2R} = 3 \alpha_s$$

$$\beta_s = \Delta_s - \alpha_s = 3 \alpha_s - \alpha_s = 2\alpha_s = \frac{L}{3R}$$

Elements of Transition curve



End point of the transition curve or starting point of the circular curve

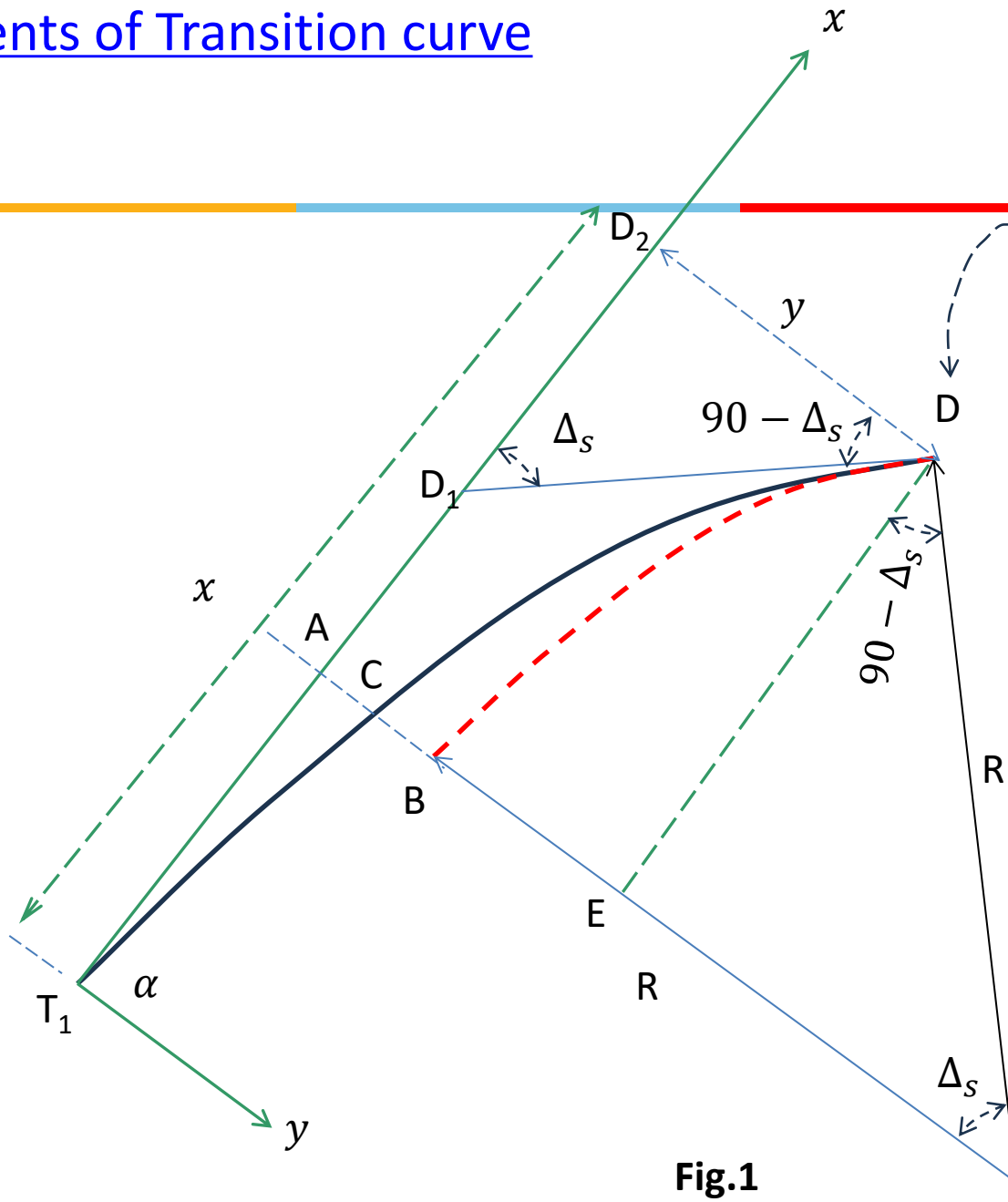
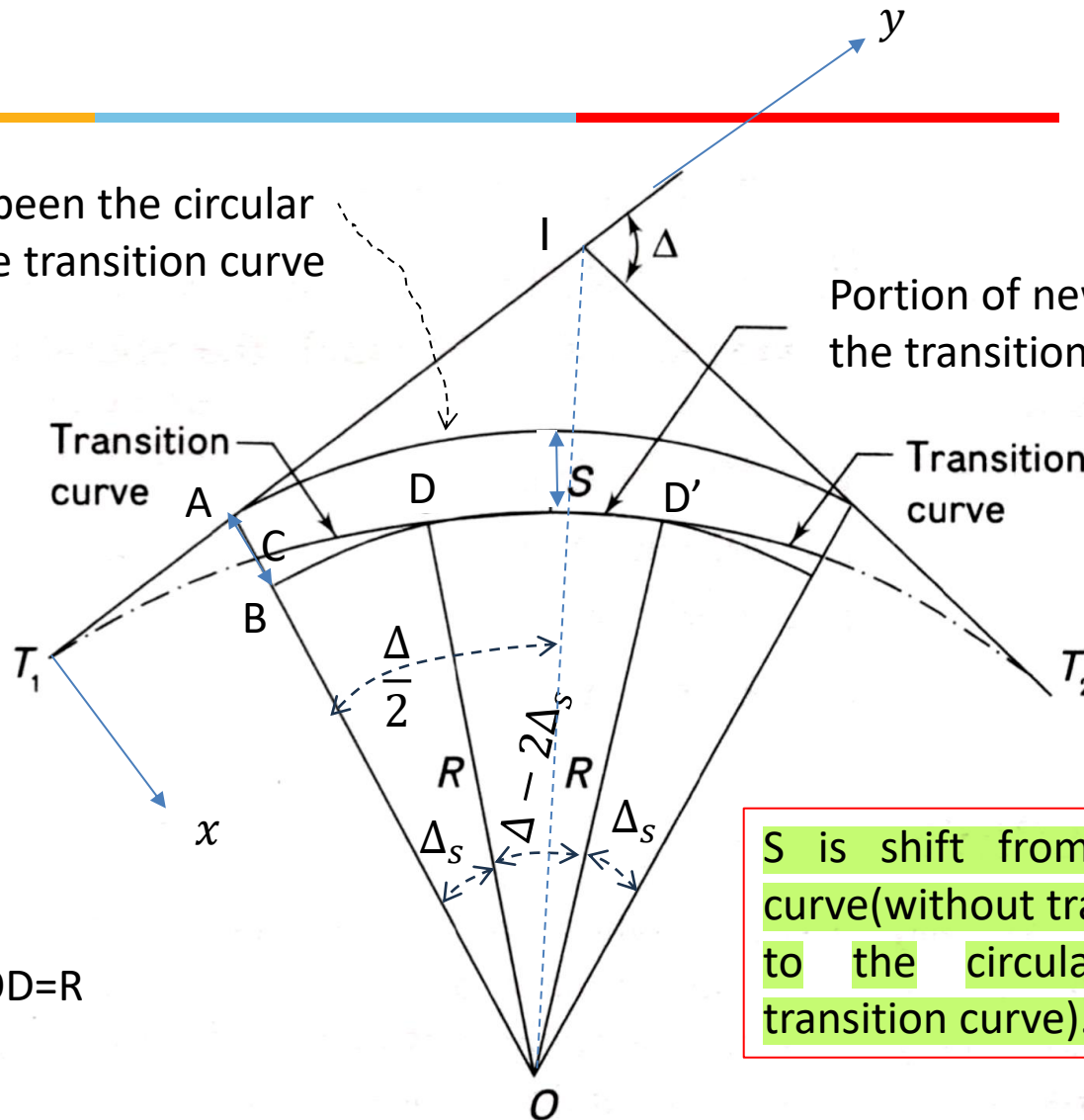


Fig.1

Portion of new circular curve with the transition curves


$$AB=S, OB=R, OD=R$$

S is shift from the circular curve (without transition curve) to the circular curve (with transition curve).

Fig.2

We have to correlate Fig.1 and Fig.2 in the following explanations: (Fig.1 is part of Fig.2)

{ Point C is the point in the transition curve, from which if we extend a line in the radial direction, we shall meet the point of curve(P.C.) of the circular curve only without the transition curves at both the ends circular curve }

DB- extended circular curve which is not existing

DC- is the portion of the transition curve which exists

Arc BD = $R\Delta_s = R \frac{L}{2R} = \frac{L}{2}$, length of circular curve BD is half of the length of transition curve

In triangle IOA(Fig.2),

$$\tan \frac{\Delta}{2} = \frac{IA}{OA} = \frac{IA}{OB + AB} = \frac{IA}{R + S}$$

$$IA = \left(\tan \frac{\Delta}{2} \right) (R + S)$$

DC is nearly equal to BD, So, $DC \cong BD = \frac{L}{2}$, Shift bisects the transition curve at point C.

(i) Shift(S)

$$S = AB = EA - EB \quad (\text{To Refer Fig.1})$$

$$S = y - (OB - OE) \quad \text{here, } y \text{ is point D}$$

$$S = y - (R - R \cos \Delta_s)$$

$$S = \frac{L^2}{6R} - R(1 - \cos \Delta_s)$$

$$S = \frac{L^2}{6R} - R \left(2 \sin^2 \frac{\Delta_s}{2} \right)$$

$$S = \frac{L^2}{6R} - 2R \left(\sin^2 \frac{\Delta_s}{2} \right)$$

$$S = \frac{L^2}{6R} - 2R \left(\frac{\Delta_s}{2} \right)^2 \quad \text{as} \quad \sin \frac{\Delta_s}{2} \cong \frac{\Delta_s}{2}$$

$$S = \frac{L^2}{6R} - \frac{R(\Delta_s)^2}{2} = \frac{L^2}{6R} - \frac{R}{2} \cdot \frac{L^2}{4R^2} = \frac{L^2}{6R} - \frac{L^2}{8R} = \frac{L^2}{24R} \quad \text{as } \Delta_s = \frac{L}{2R}$$

$$S = \frac{L^2}{24R}$$

(ii) Tangent Length(IT_1) (To refer Fig. 2 and Fig. 1)

$$IT_1 = IA + AT_1$$

$$= (R + S) \left(\tan \frac{\Delta}{2} \right) + (T_1 D_2 - AD_2)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + (x - R \sin \Delta_s), \quad x \text{ is distance of point } D \text{ along the tangent}$$

(i) For true spiral:

$$x = l \left(1 - \frac{\phi^2}{10} \right)$$

But when x becomes the distance of point D along the tangent, we can say,

$$x = L \left(1 - \frac{(\Delta_s)^2}{10} \right)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + (x - R \sin \Delta_s) = (R + S) \left(\tan \frac{\Delta}{2} \right) + \left(L \left(1 - \frac{(\Delta_s)^2}{10} \right) - R \sin \Delta_s \right)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \left(L \left(1 - \frac{(\Delta_s)^2}{10} \right) - R \sin \Delta_s \right)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \left(L \left(1 - \frac{(L)^2}{40R^2} \right) - R \left(\Delta_s - \frac{(\Delta_s)^3}{3!} \right) \right) \text{ as } \sin \Delta_s = \Delta_s - \frac{(\Delta_s)^3}{3!}$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \left(L \left(1 - \frac{(L)^2}{40R^2} \right) - R \left(\frac{L}{2R} - \frac{L^3}{48R^3} \right) \right) \text{ as } \Delta_s = \frac{L}{2R}$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \left(L - \frac{L^3}{40R^2} - \frac{L}{2} + \frac{L^3}{48R^2} \right)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \left(\frac{L}{2} - \left(\frac{L^3}{40R^2} - \frac{L^3}{48R^2} \right) \right)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \left(\frac{L}{2} - \left(\frac{6L^3 - 5L^3}{240R^2} \right) \right)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \left(\frac{L}{2} - \frac{L^3}{240R^2} \right)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \frac{L}{2} \left(1 - \frac{L^2}{120R^2} \right)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \frac{L}{2} \left(1 - \frac{S}{24R} \right) \text{ as } S = \frac{L^2}{24R}$$

(ii) For cubic spiral: (in cubic spiral, $x \cong L$, where x is distance of point D along the tangent)

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + (x - R \sin \Delta_s)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + (L - R \Delta_s) , \text{ as } \sin \Delta_s \cong \Delta_s$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \left(L - R \frac{L}{2R} \right) , \text{ as } \Delta_s = \frac{L}{2R}$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \left(L - \frac{L}{2} \right)$$

$$IT_1 = (R + S) \left(\tan \frac{\Delta}{2} \right) + \frac{L}{2}$$

(iii) Total length of curve

$$\begin{aligned}
 \text{Total length of curve} &= T_1D + DD' + D'T_2 = L + R(\Delta - 2\Delta_s) + L \\
 &= L + R\left(\Delta - 2\frac{L}{2R}\right) + L \\
 &= L + R\left(\Delta - \frac{L}{R}\right) + L, \quad \Delta \text{ in radian}
 \end{aligned}$$

(iv) y co-ordinate of point C

For a cubic parabola, at point C, $x = \frac{L}{2}$

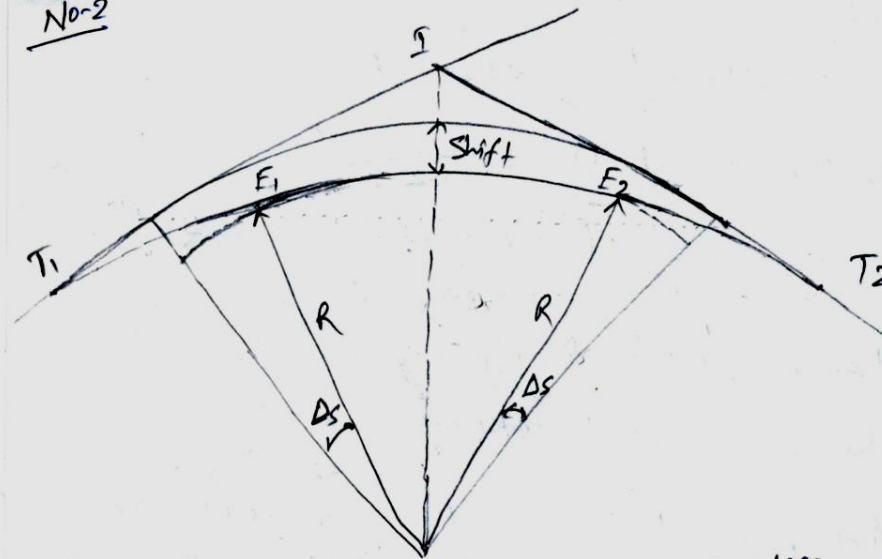
$$\text{So, at point C, } y = \frac{x^3}{6RL} = \frac{\left(\frac{L}{2}\right)^3}{6RL} = \frac{L^2}{48R} = \frac{S}{2}$$

The transition curve bisects the shift at point C. Point C is one of the main points of interest.

Q8: A road 7m wide is to deflect through an angle of 45° with centerline radius of 350m. The chainage of the intersection point is 3350m. A circular arc is to be provided with transition curves in the form a **cubic parabola** at the ends of circular curves, of such lengths that the rate of change of radial acceleration can be maintained as 0.45m/s^2 per second. The speed of the vehicle is 60 Km/hour. Find out the followings,

- (a) Length of transition curves and circular curve
- (b) Superelevation
- (c) Chainage of all junction points
- (d) Find out the offsets of transition curves at an interval of 8m.

No-2



Speed of vehicle = $V = 60 \text{ km/hour} = \frac{60 \times 1000}{3600} = 16.666 \text{ m/sec}$

$\Delta = 45^\circ$ (Angle of deflection) = $\frac{\pi}{4} = 0.785 \text{ radian}$.

$R = \text{Radius of circular curve} = 350 \text{ m}$

$B = \text{Width of road} = 7 \text{ m}$

$\alpha = \text{rate of change of radial acc per second} = 0.45$

(a) Length of transition curve and circular curve

$T_1 E_1 = L = \text{Length of transition curve} = \frac{V^3}{R \alpha} = \frac{(16.666)^3}{350 \times 0.45} = 29.39 \text{ m}$ ①

$\Delta_s = \text{Spiral Angle} = \frac{L}{2R} = \frac{29.39}{2 \times 350} = 0.0419 \text{ radian}$
 $= 2.405 \text{ degree}$

$E_1 E_2 = \text{Length of Circular Curve}$
 $= R(\Delta - 2\Delta_s) = 350(0.785 - 2 \times 0.0419) = 245.42 \text{ m}$ ①

(b) Superelevation. $h = \frac{BV^2}{gR} = \frac{7 \times (16.666)^2}{9.81 \times 350} = 0.566 \text{ m}$ ②

(c) Chainage of All junctions.

$$S = \text{shift} = \frac{L^2}{24R} = 0.102 \text{ m}$$

$$\begin{aligned} \text{Length of tangent} &= (R+S) \tan \frac{A}{2} + \frac{L}{2} \\ &= (350 + 0.102) \tan \frac{75}{2} + \frac{29.39}{2} \\ &= 159.7119 \text{ m} \end{aligned}$$

(2) Chainage of $T_1 = 3350 - 159.7119 = 3190.288 \text{ m}$ Length of transition curve

Chainage of 1st junction point (E_1) = $3190.288 + 29.39$

$$= 3219.678 \text{ m}$$

Chainage of 2nd junction point = $3219.678 + 245.42$

$$= 3465.098 \text{ m}$$

Chainage of $T_2 = 3465.098 + 29.39$ (E, F_2)

~~Length of~~ $= 3494.488 \text{ m}$

(d) Offsets at an interval of 8m

At 0 it is 0

At 8m it is 0.00829m

At 16m it is 0.0663m

At 24m it is 0.2239m

At 29.39m it is ~~0.4113m~~ 0.4113m

$$y = \frac{x^3}{6LR}$$

